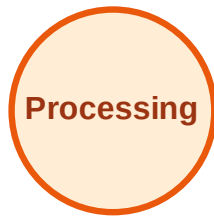


BFS Traversal

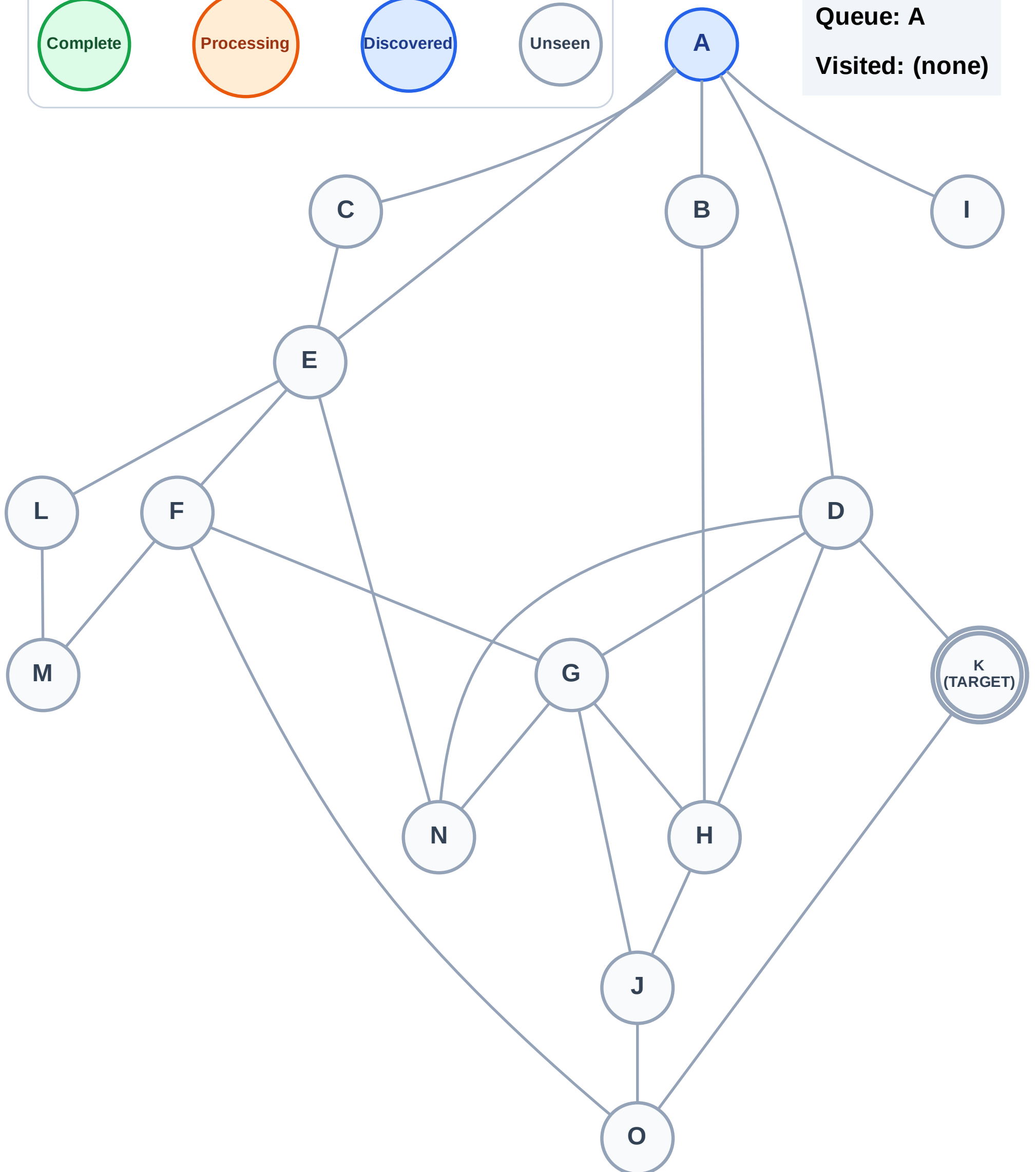
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

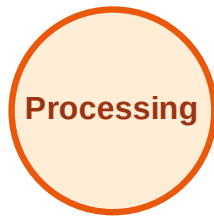
Visited: (none)



BFS Traversal

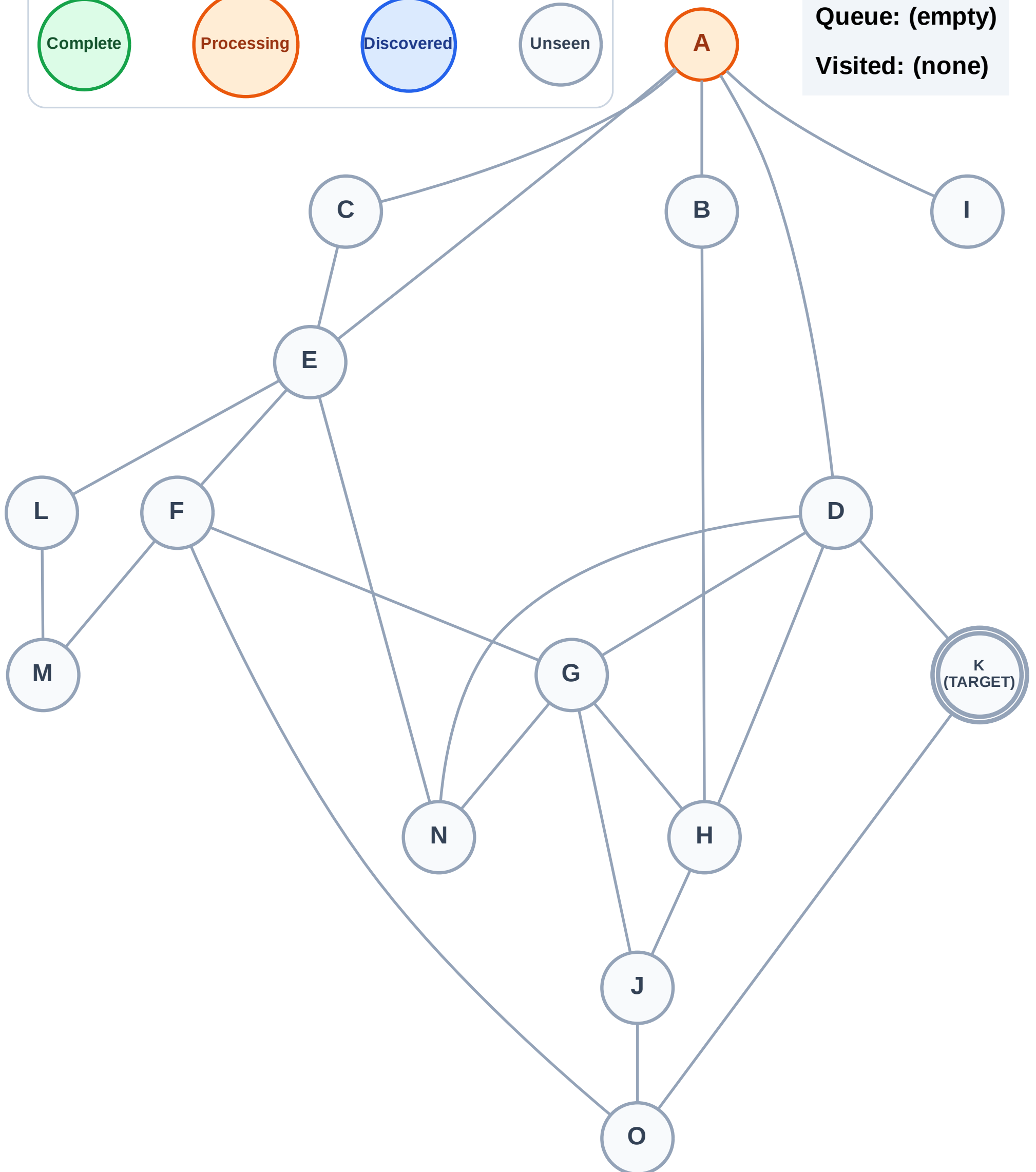
Step 2: Process node A

Legend



Queue: (empty)

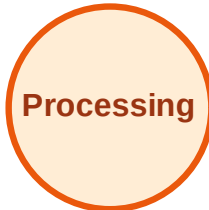
Visited: (none)



BFS Traversal

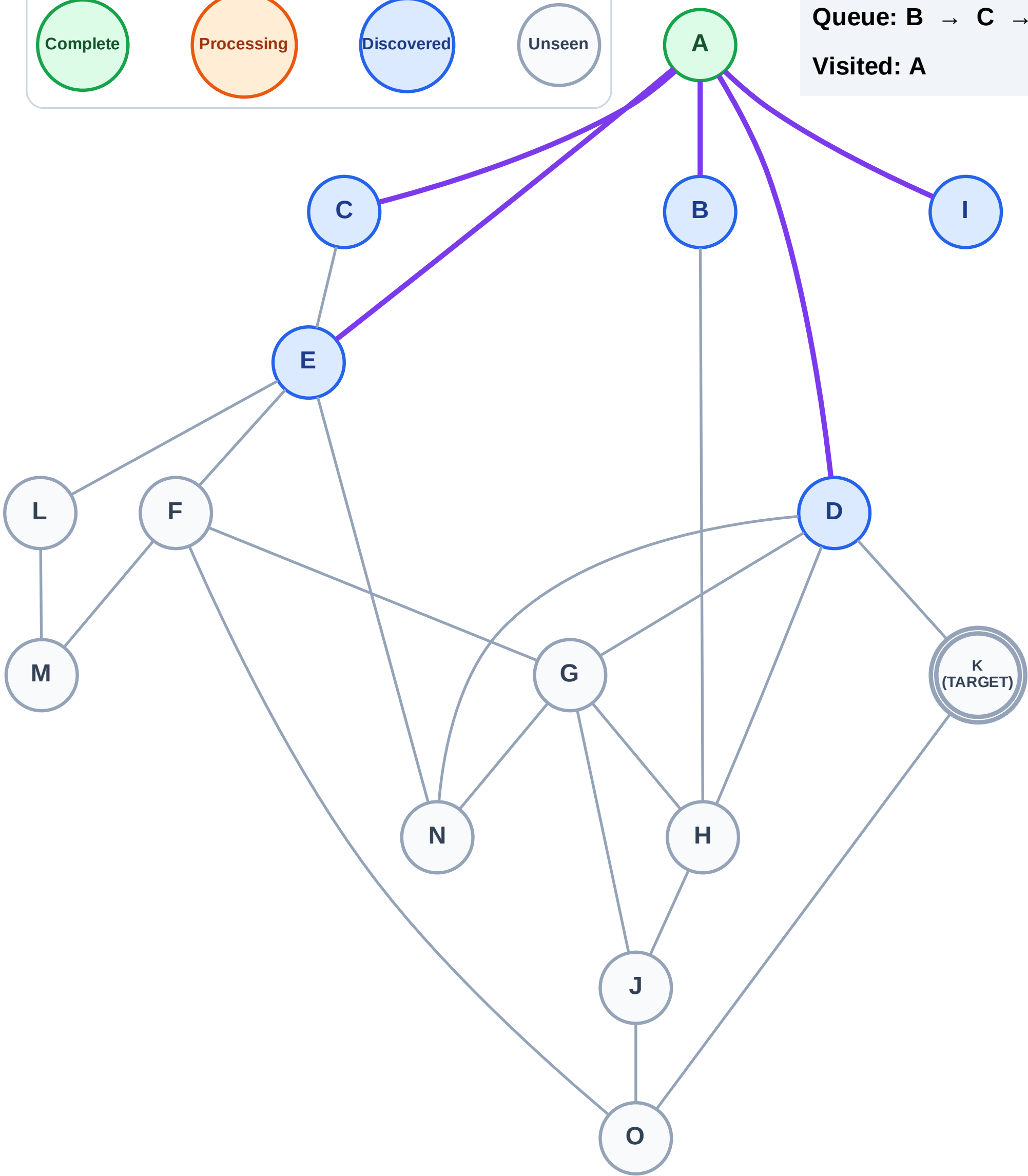
Step 3: Discover B, C, D, E, I from A

Legend



Queue: B → C → D → E → I

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

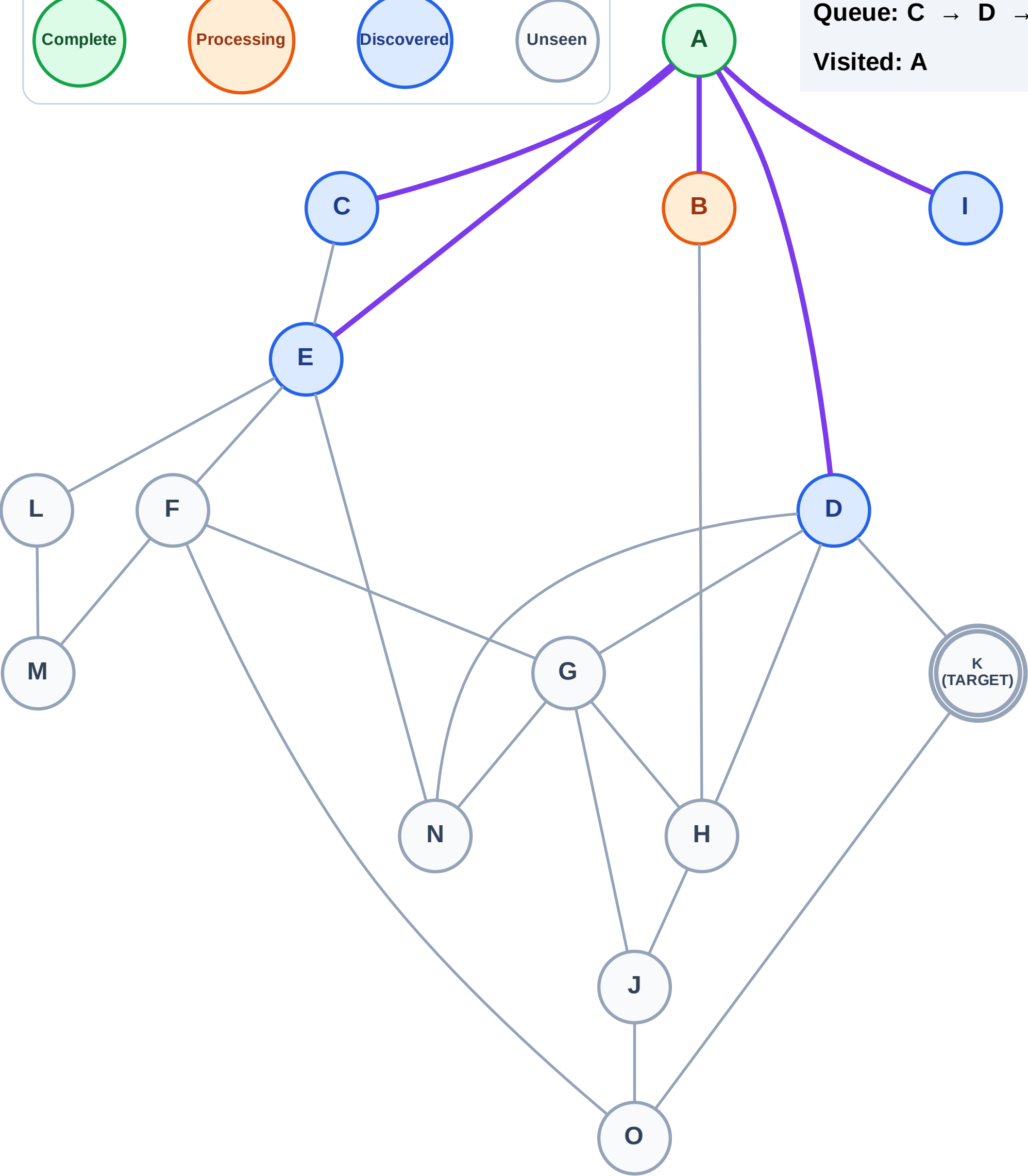
Processing

Discovered

Unseen

Queue: C → D → E → I

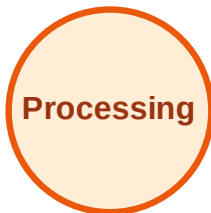
Visited: A



BFS Traversal

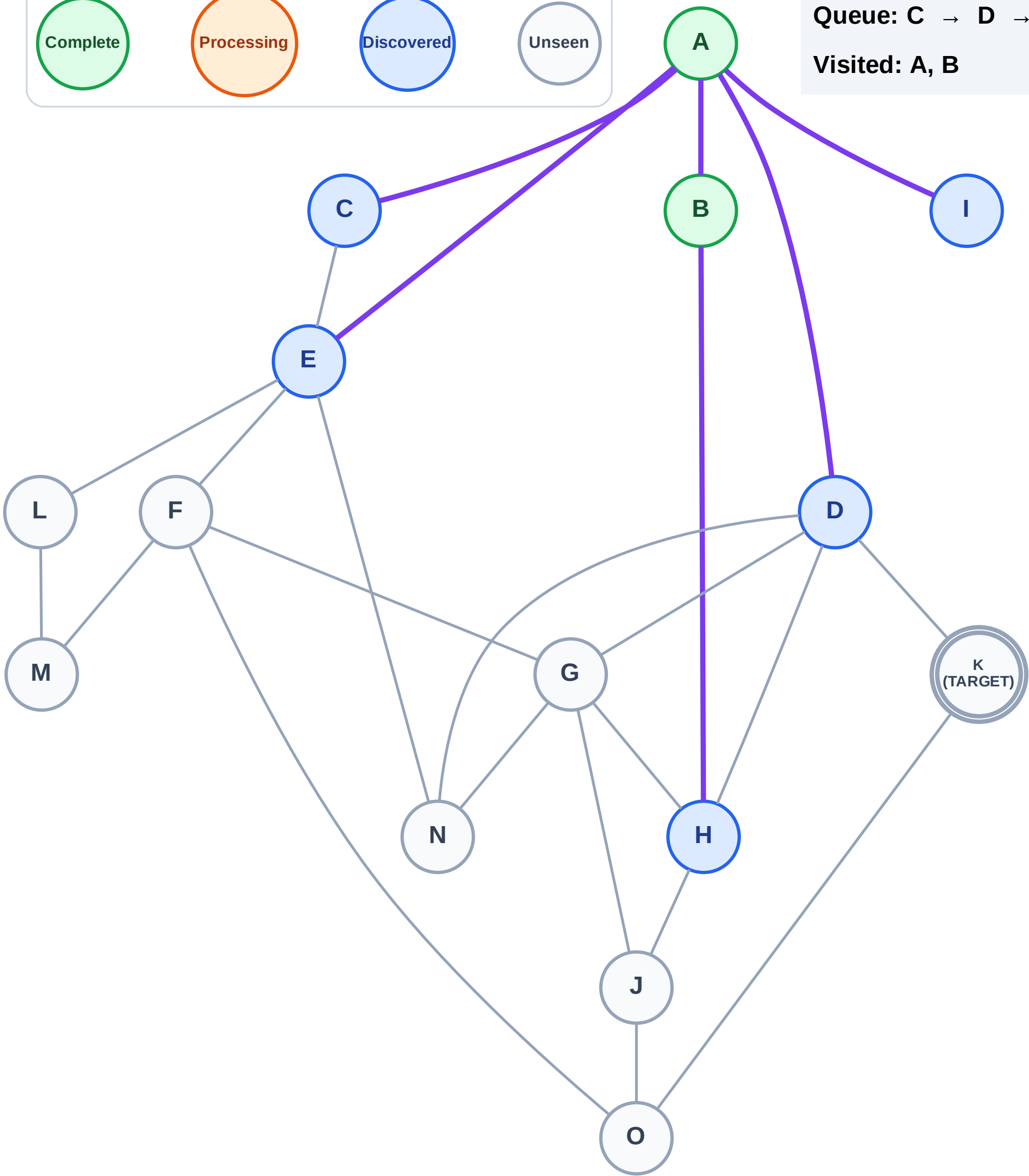
Step 5: Discover H from B

Legend



Queue: C → D → E → I → H

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

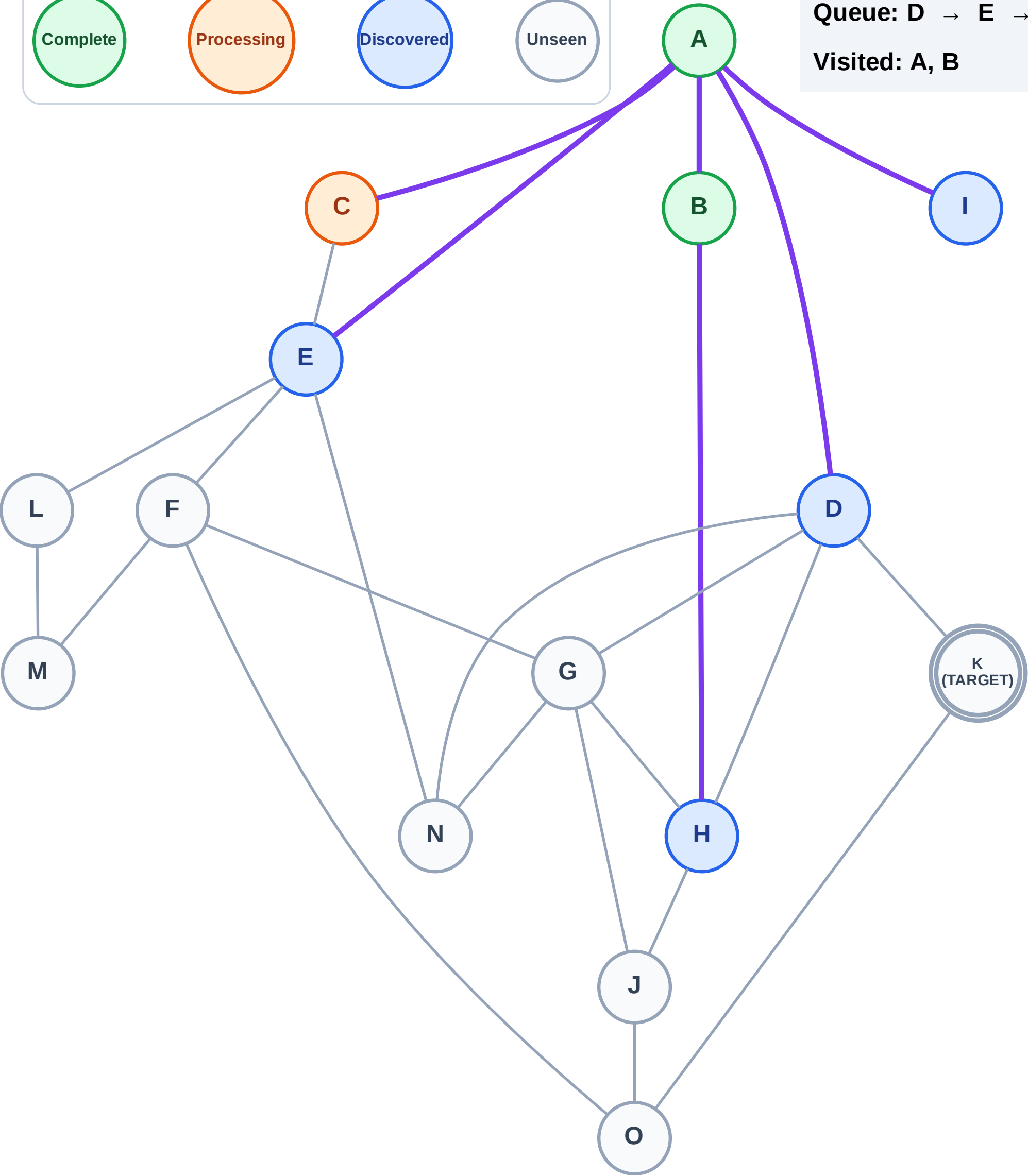
Processing

Discovered

Unseen

Queue: D → E → I → H

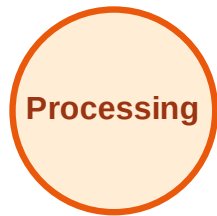
Visited: A, B



BFS Traversal

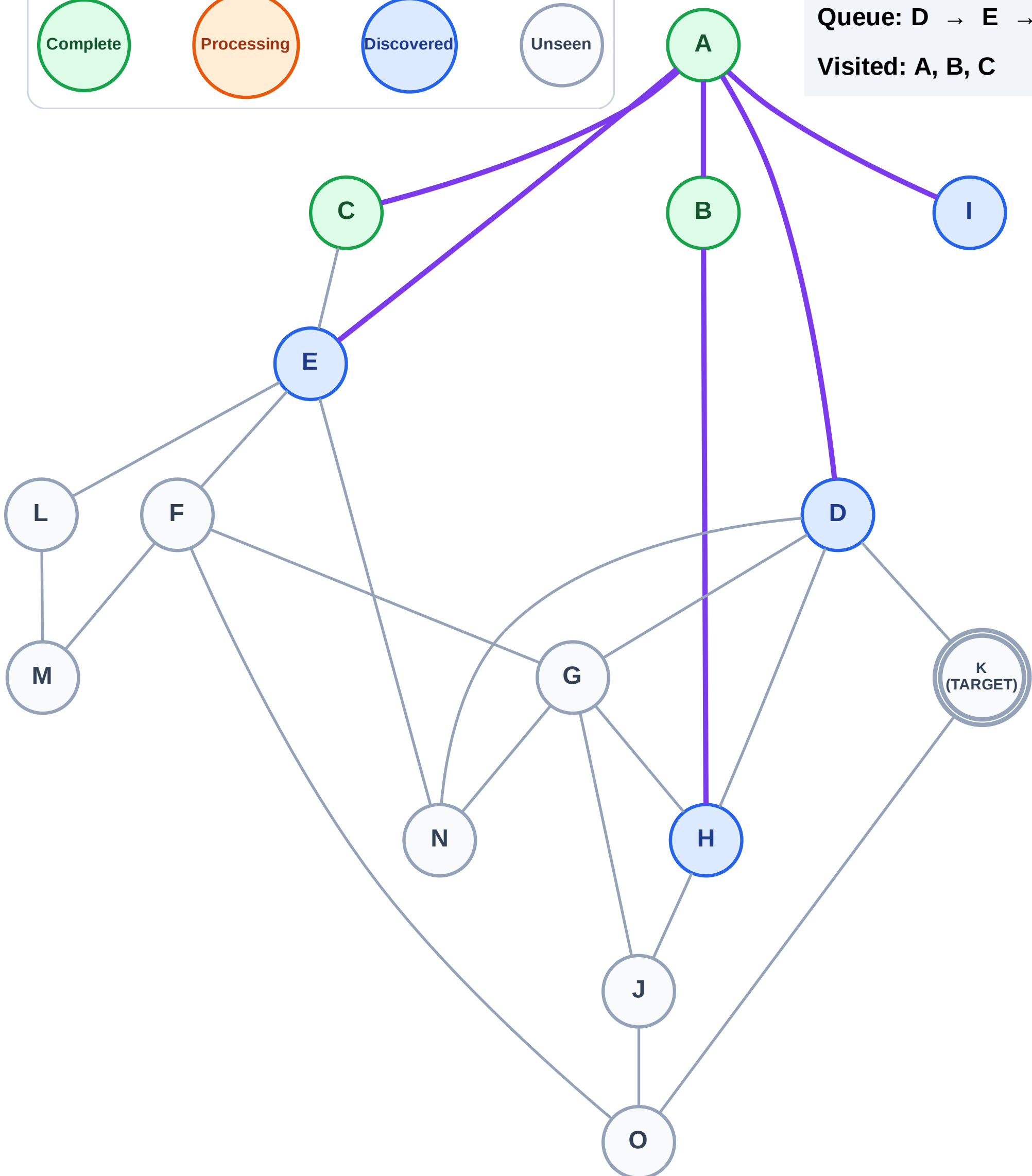
Step 7: No new neighbors from C

Legend



Queue: D → E → I → H

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

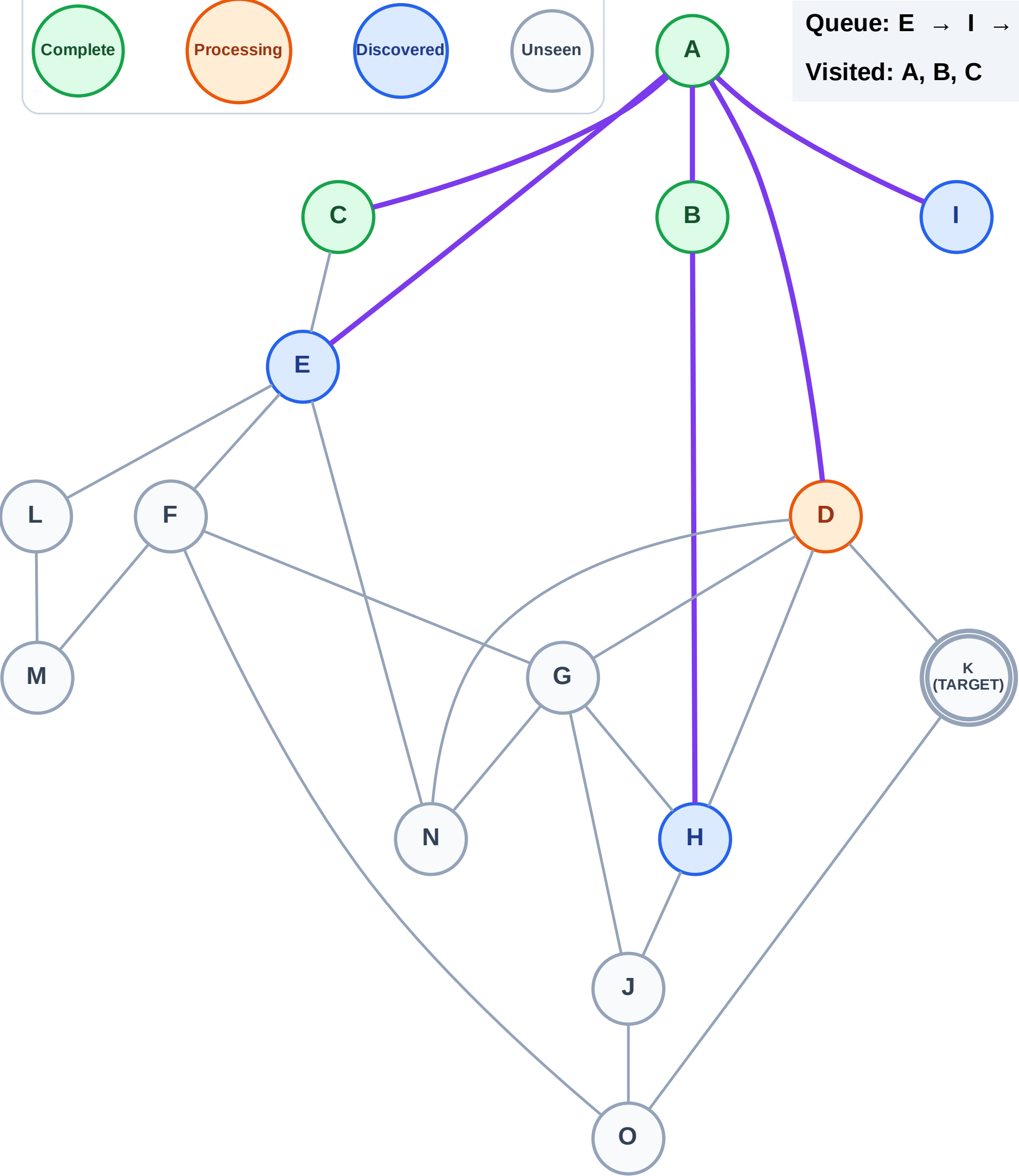
Processing

Discovered

Unseen

Queue: E → I → H

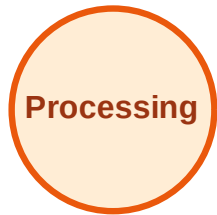
Visited: A, B, C



BFS Traversal

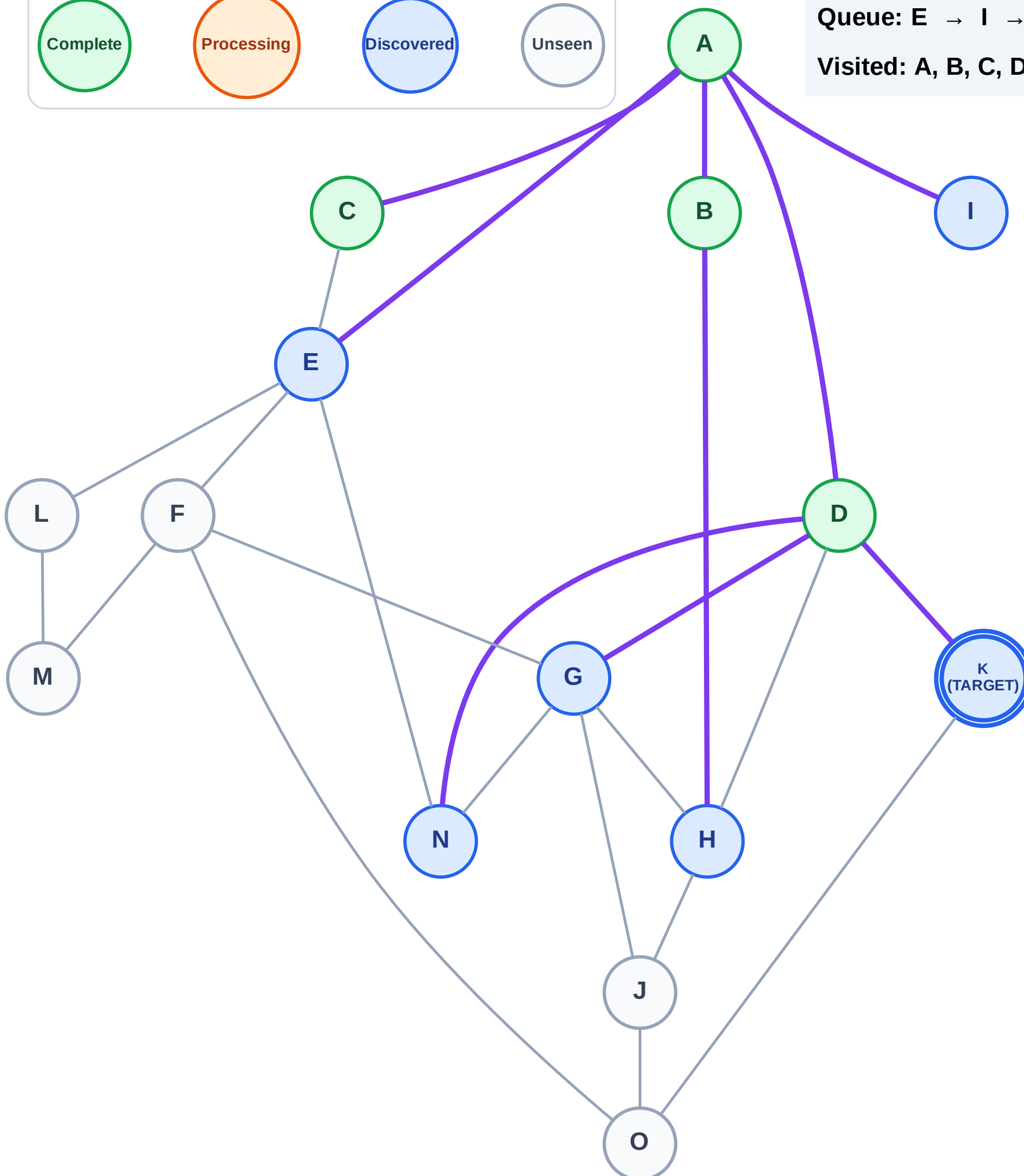
Step 9: Discover G, K, N from D

Legend



Queue: E → I → H → G → K → N

Visited: A, B, C, D



BFS Traversal

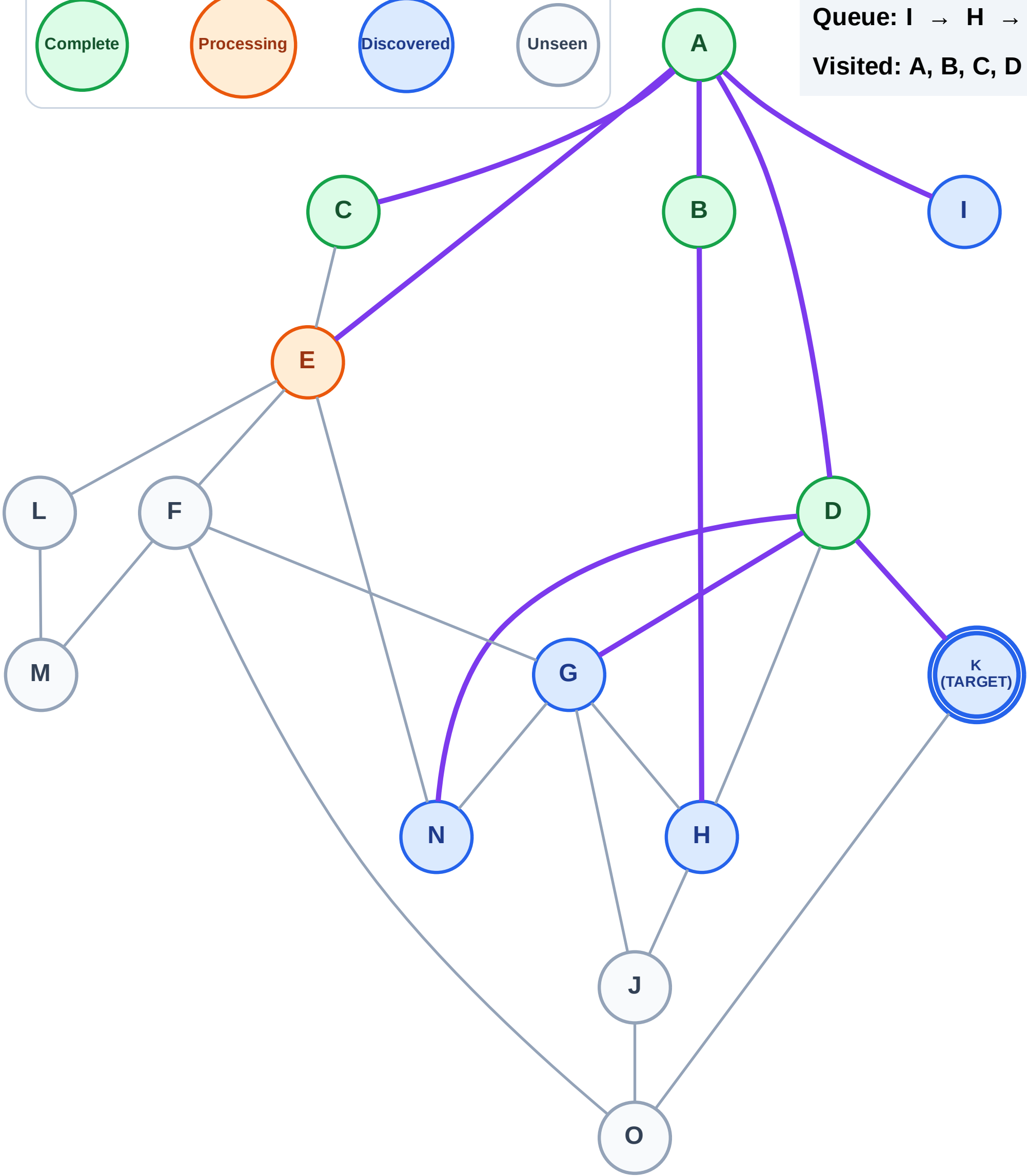
Step 10: Process node E

Legend



Queue: I → H → G → K → N

Visited: A, B, C, D



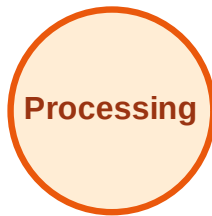
BFS Traversal

Step 11: Discover F, L from E

Legend



Complete



Processing



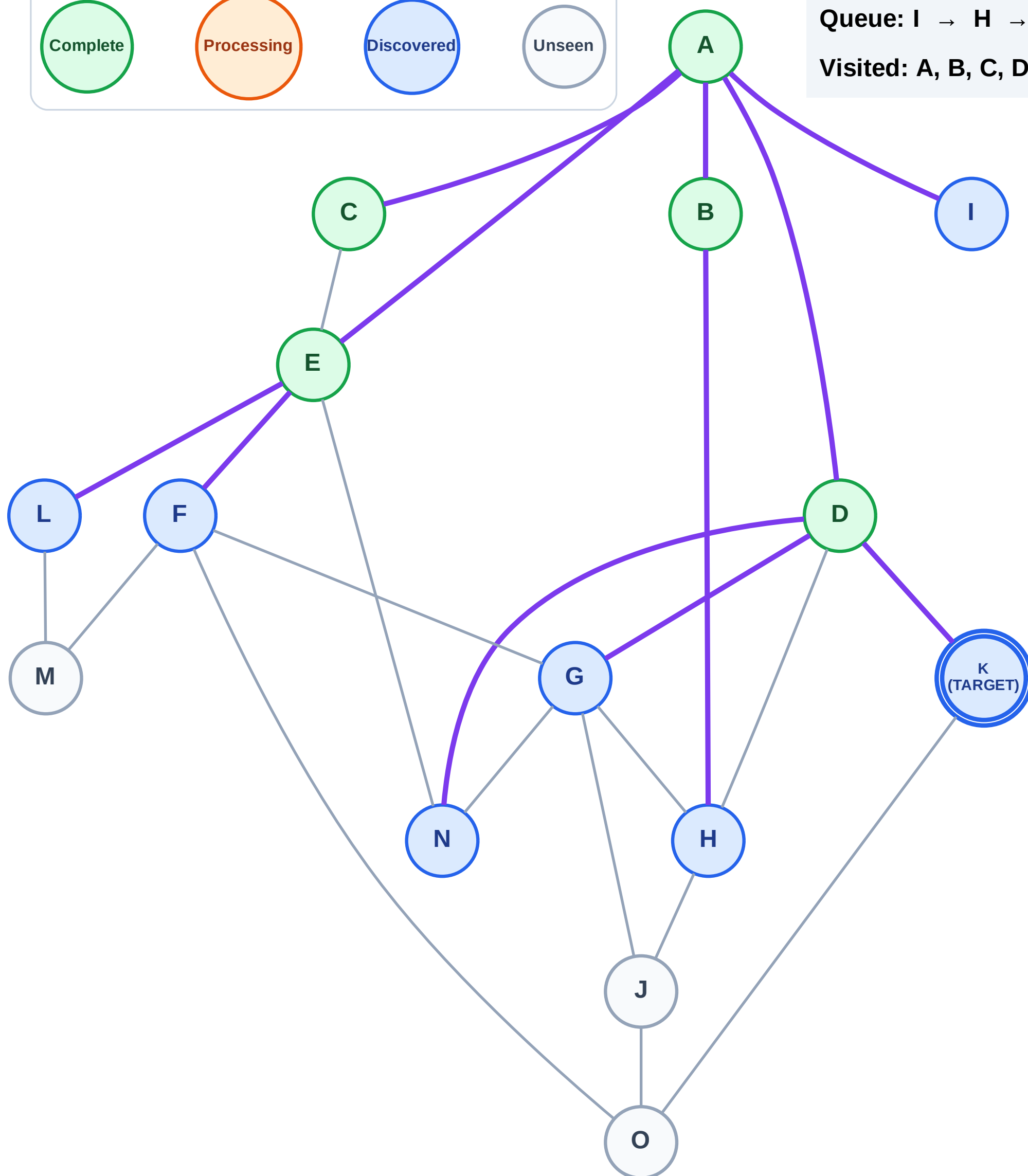
Discovered



Unseen

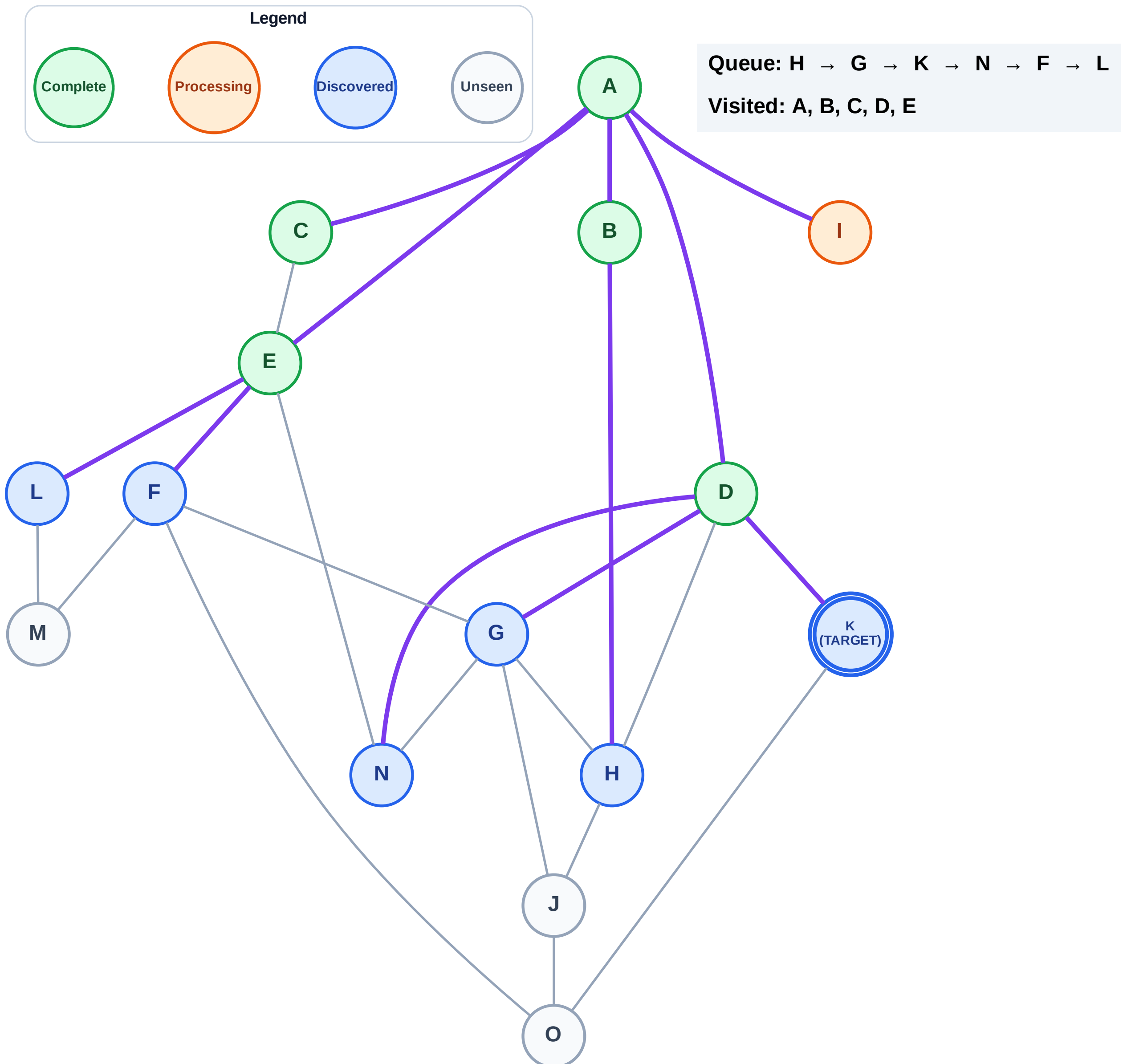
Queue: I → H → G → K → N → F → L

Visited: A, B, C, D, E



Step 12: Process node I

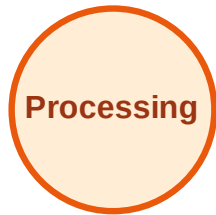
Step 12: Process node I



BFS Traversal

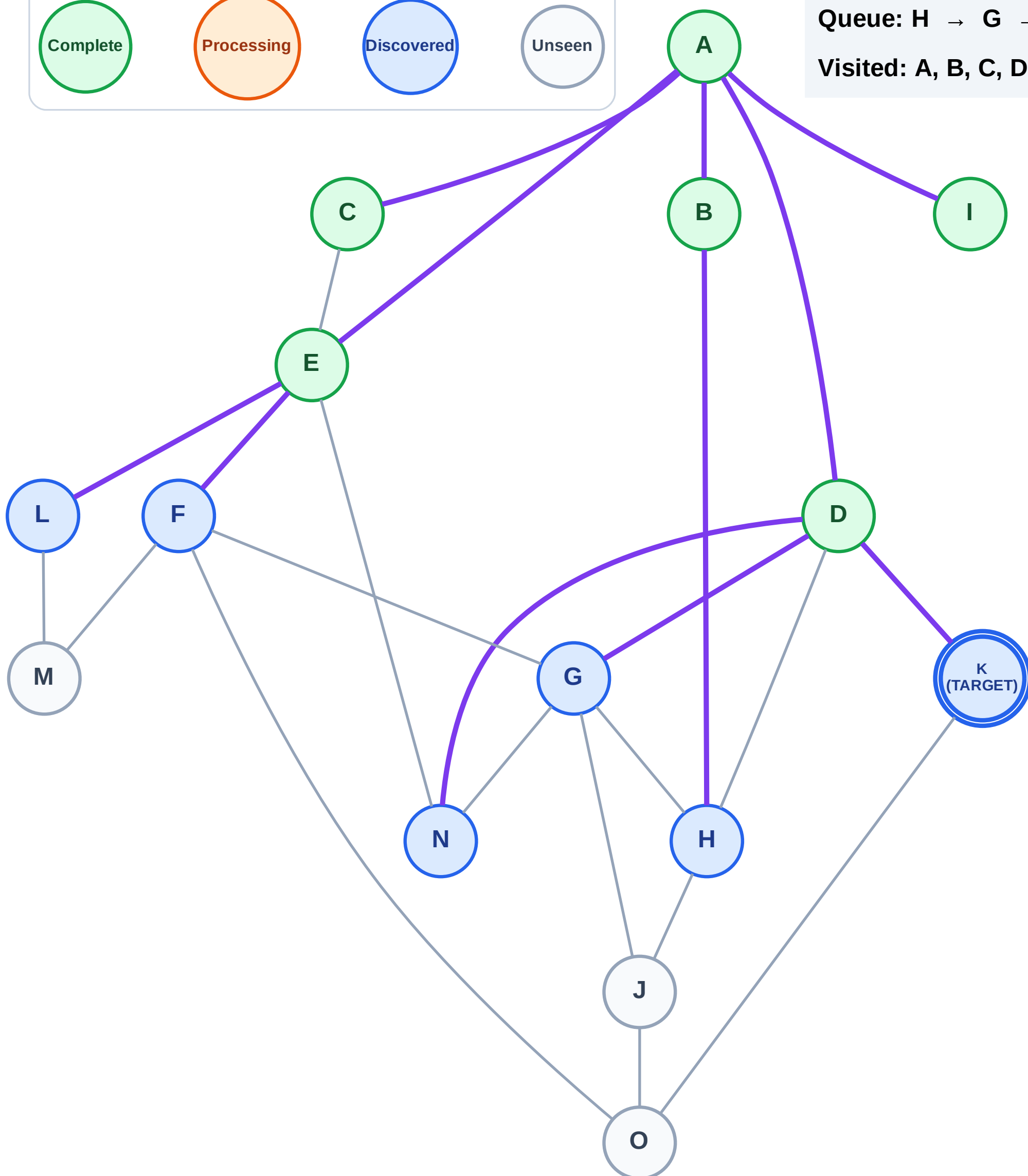
Step 13: No new neighbors from I

Legend



Queue: H → G → K → N → F → L

Visited: A, B, C, D, E, I



BFS Traversal

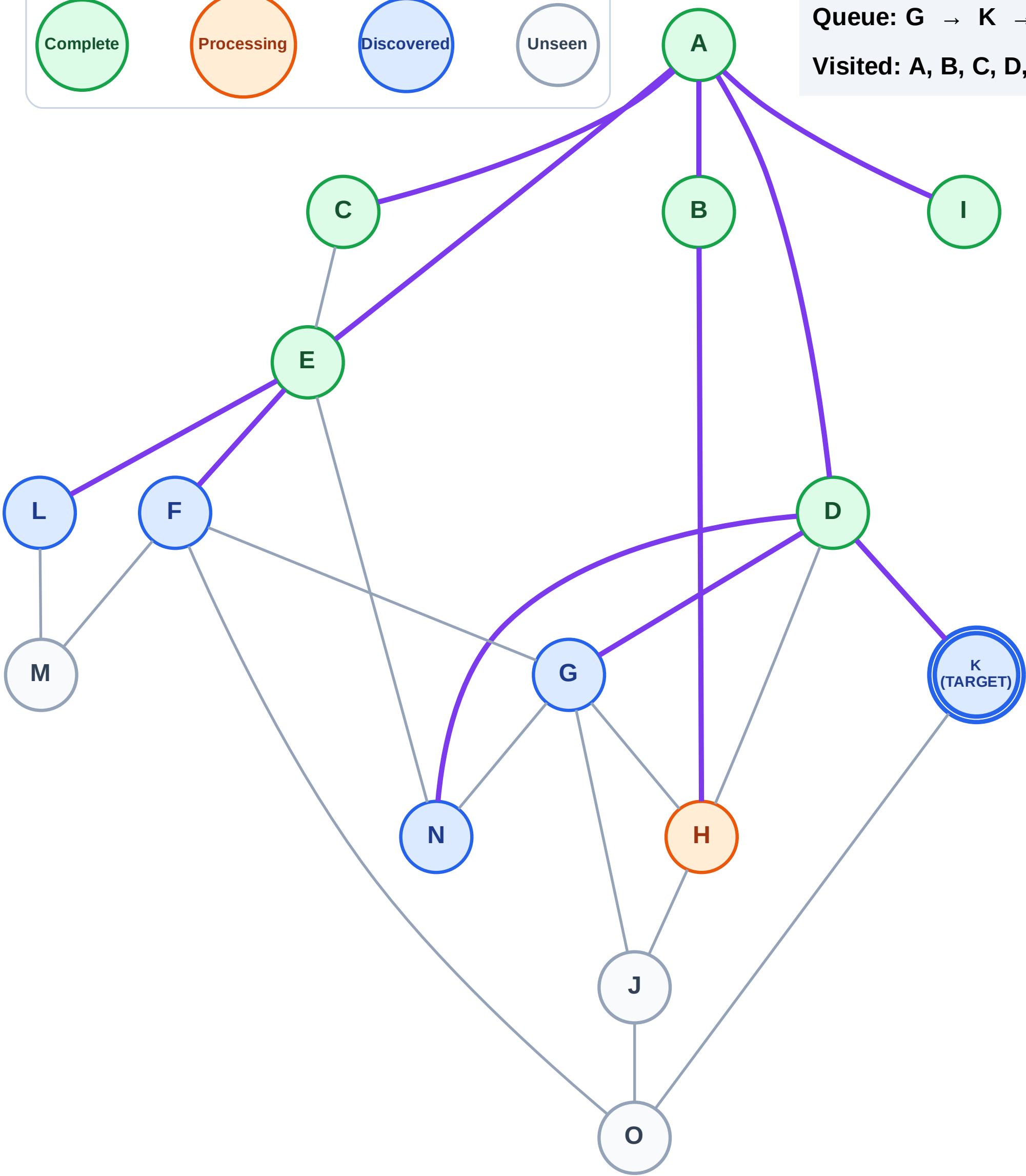
Step 14: Process node H

Legend



Queue: G → K → N → F → L

Visited: A, B, C, D, E, I



BFS Traversal

Step 15: Discover J from H

Legend

Complete

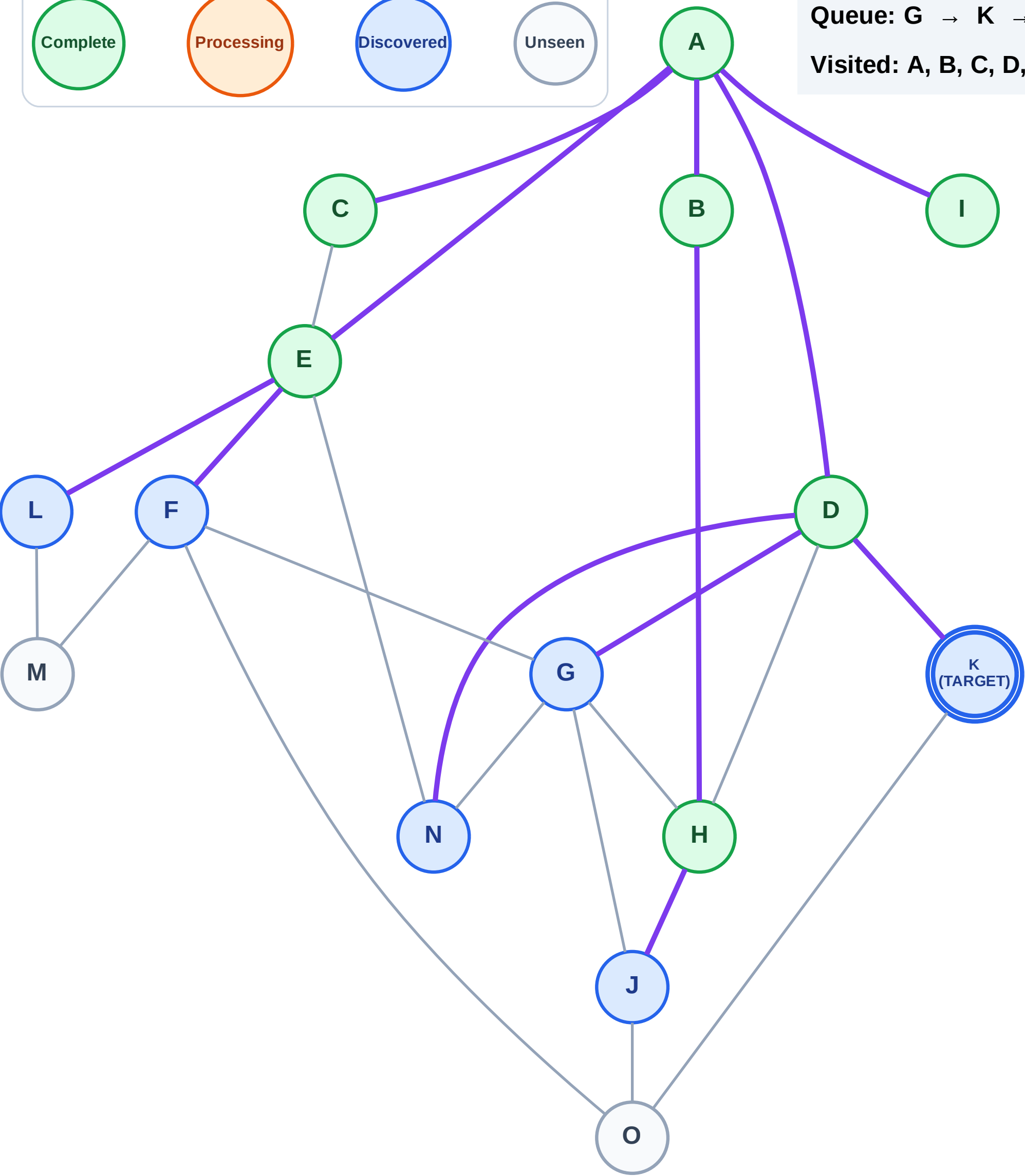
Processing

Discovered

Unseen

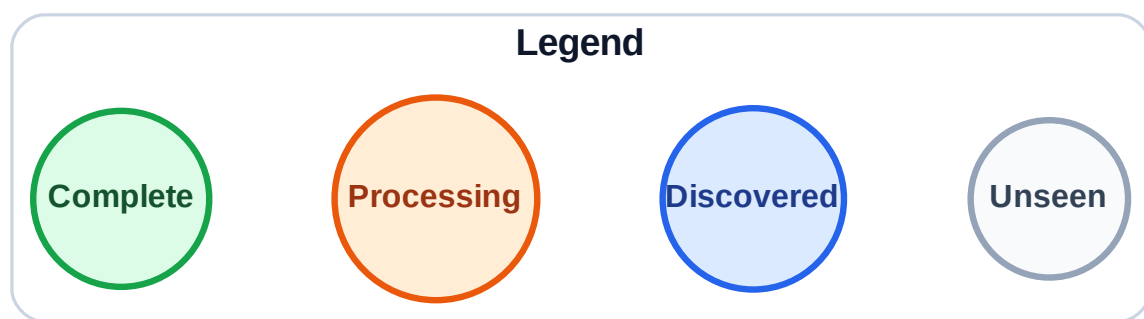
Queue: G → K → N → F → L → J

Visited: A, B, C, D, E, H, I



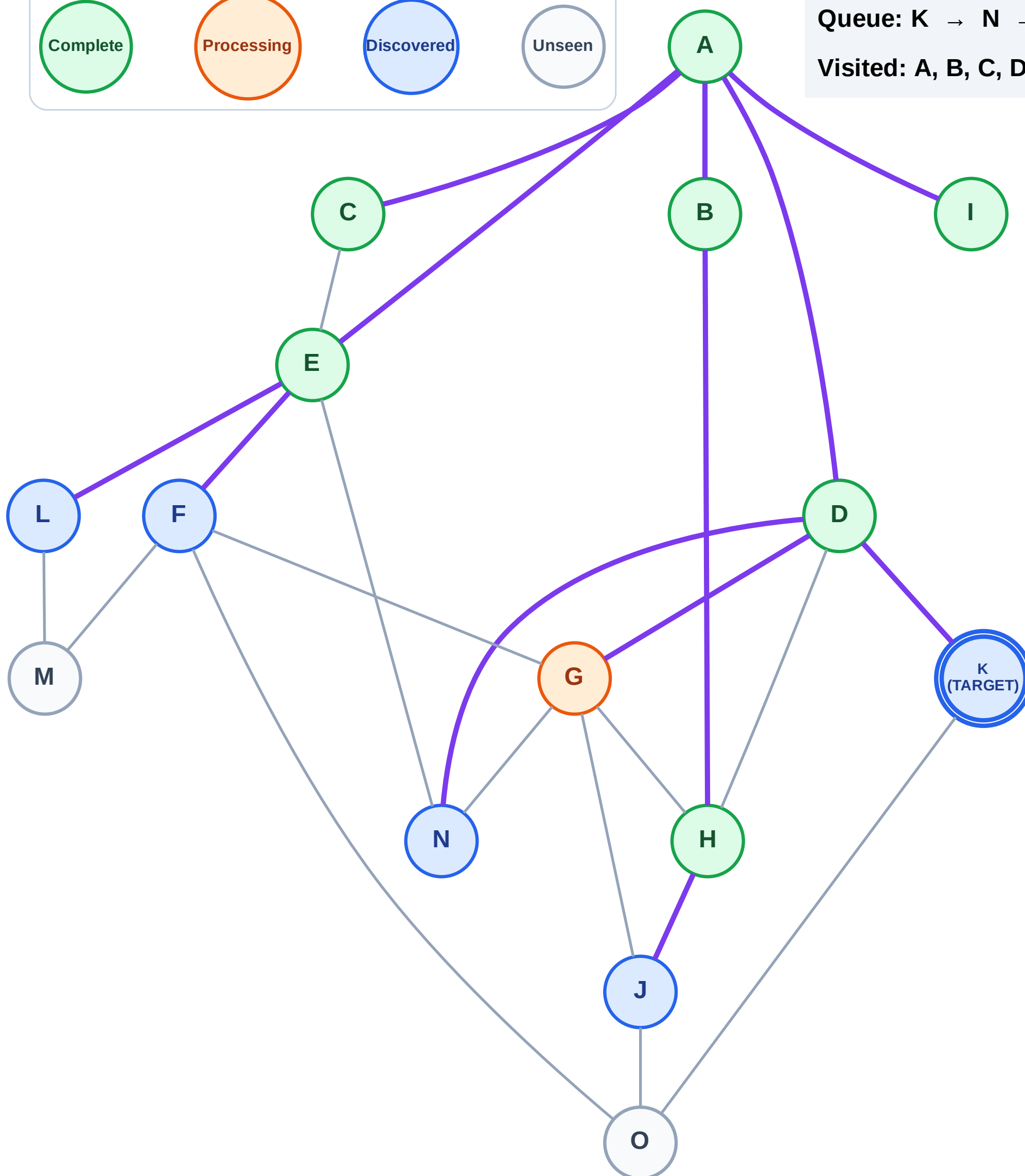
BFS Traversal

Step 16: Process node G



Queue: K → N → F → L → J

Visited: A, B, C, D, E, H, I



BFS Traversal

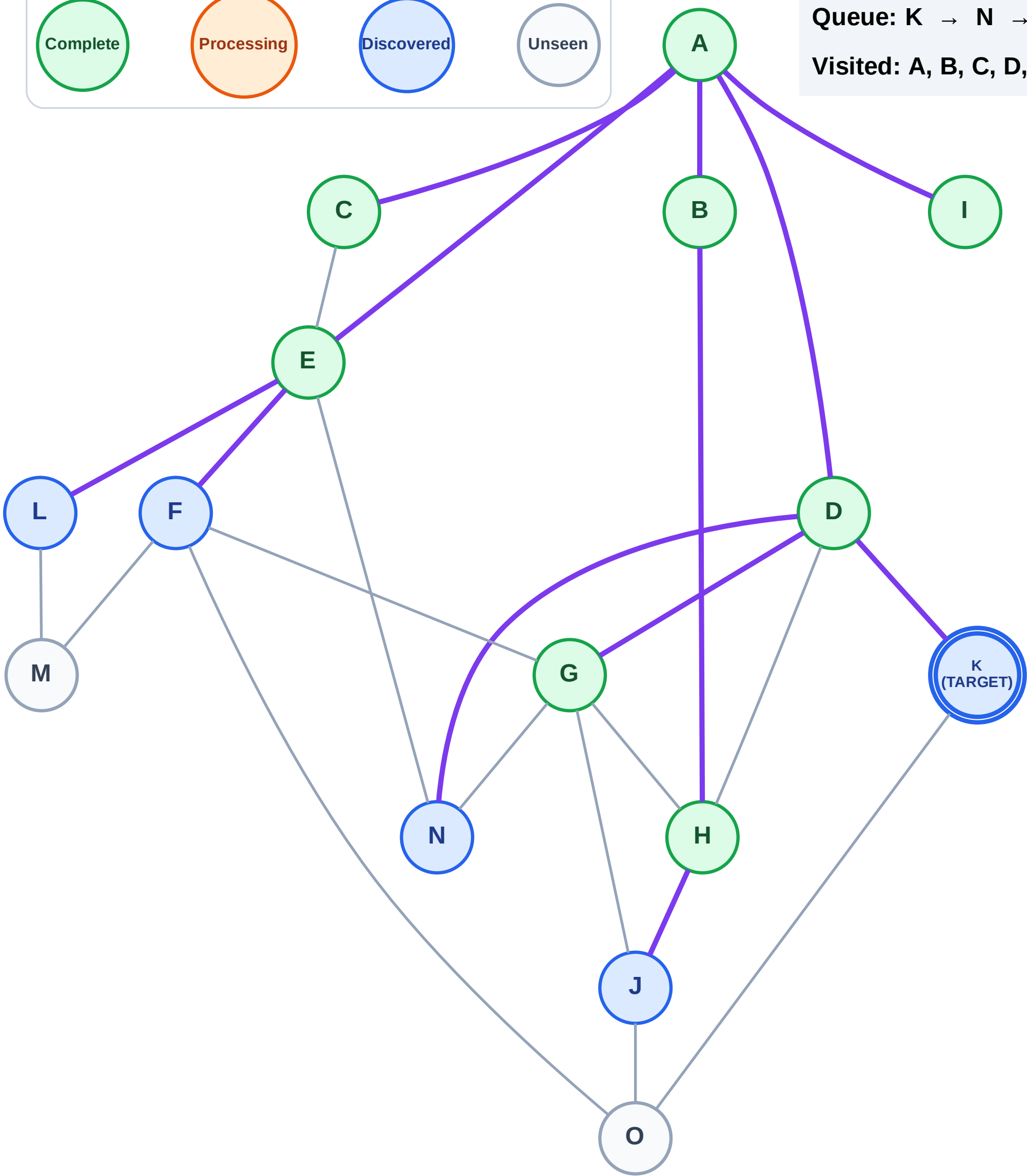
Step 17: No new neighbors from G

Legend



Queue: K → N → F → L → J

Visited: A, B, C, D, E, G, H, I



BFS Traversal

Step 18: Process node K

Legend

Complete

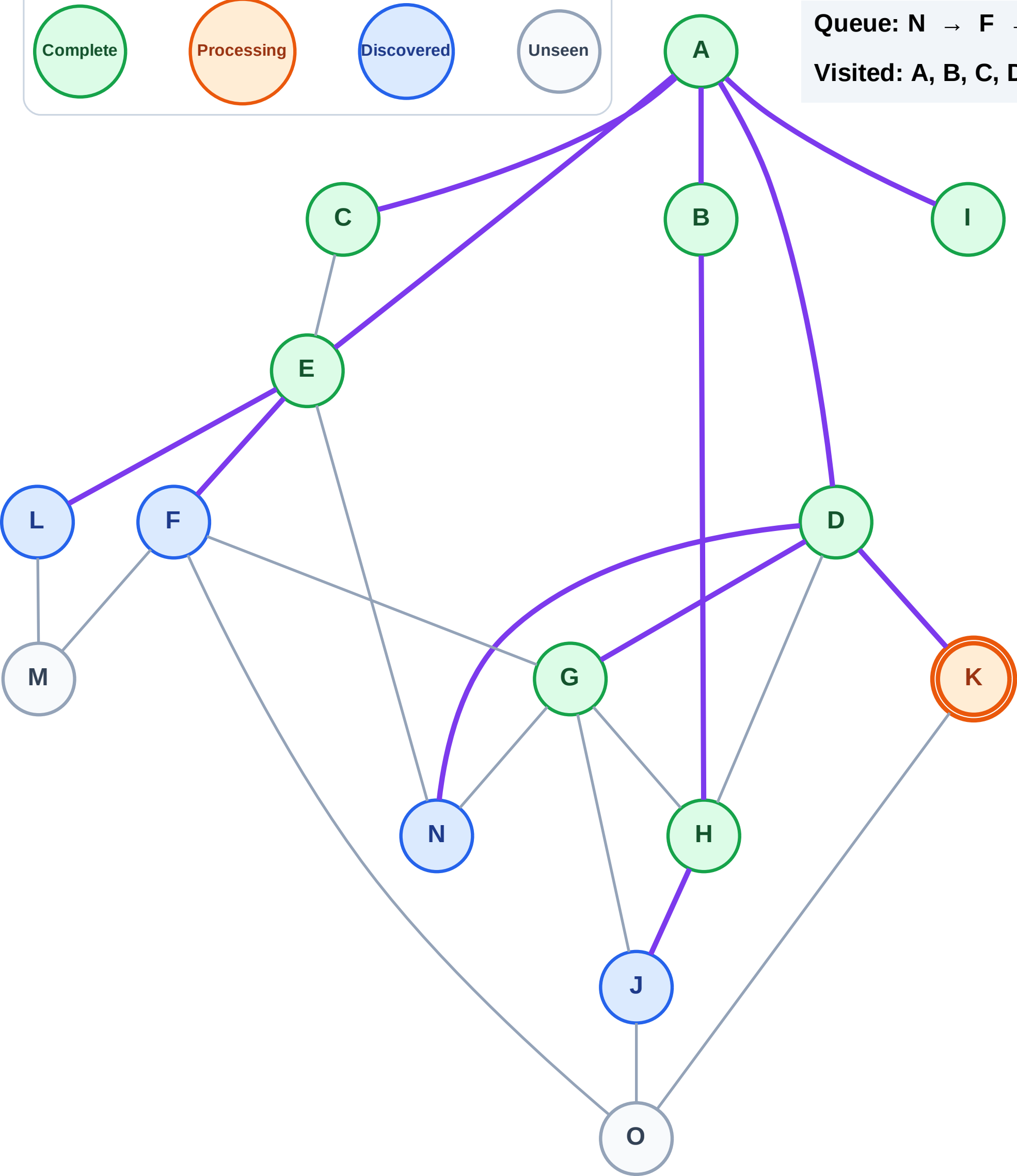
Processing

Discovered

Unseen

Queue: N → F → L → J

Visited: A, B, C, D, E, G, H, I



BFS Traversal

Step 19: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend

Complete

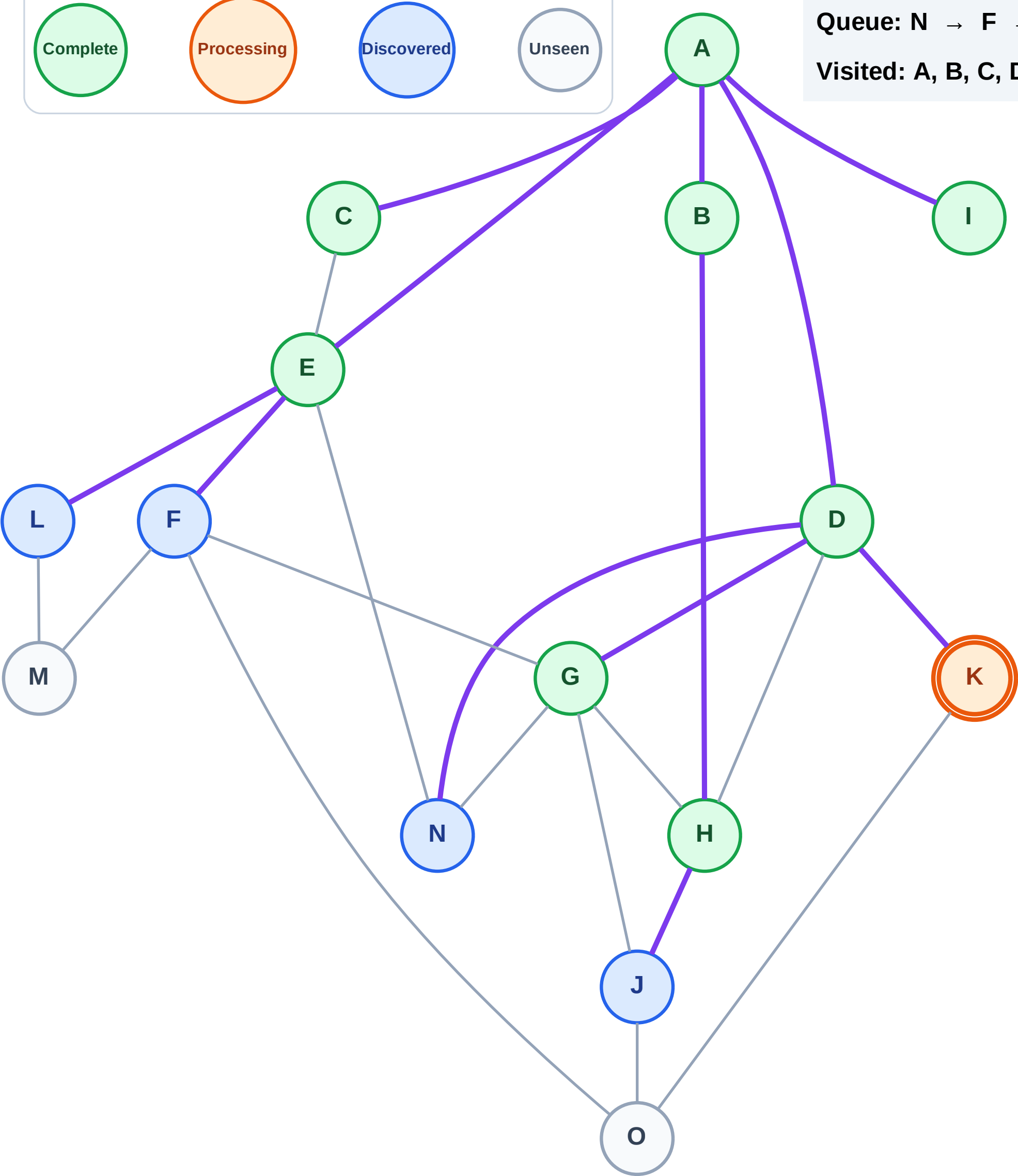
Processing

Discovered

Unseen

Queue: N → F → L → J

Visited: A, B, C, D, E, G, H, I, K



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

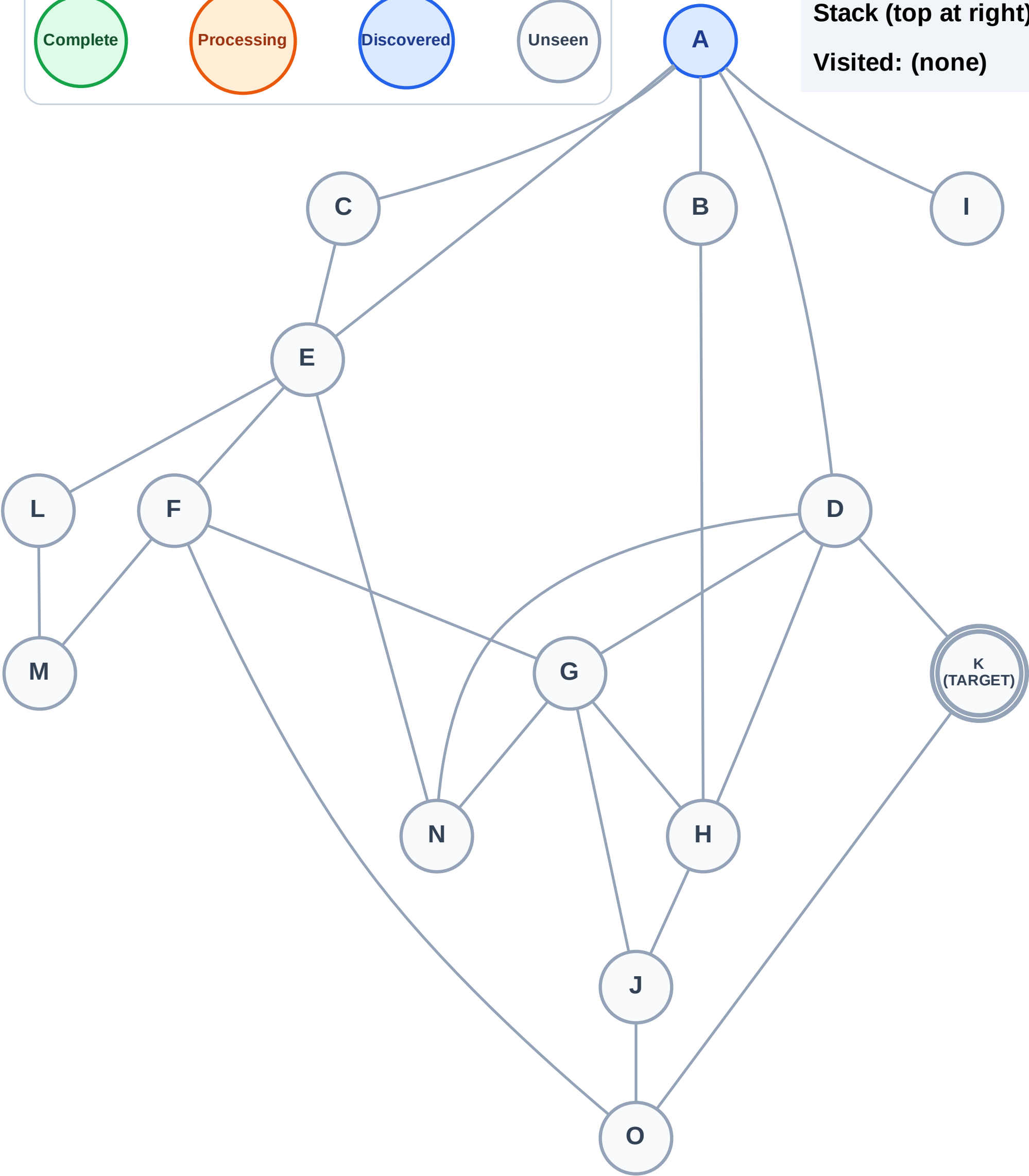
Complete

Processing

Discovered

Unseen

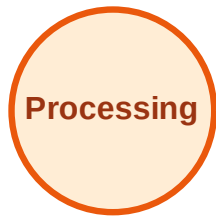
Stack (top at right): A
Visited: (none)



DFS Traversal

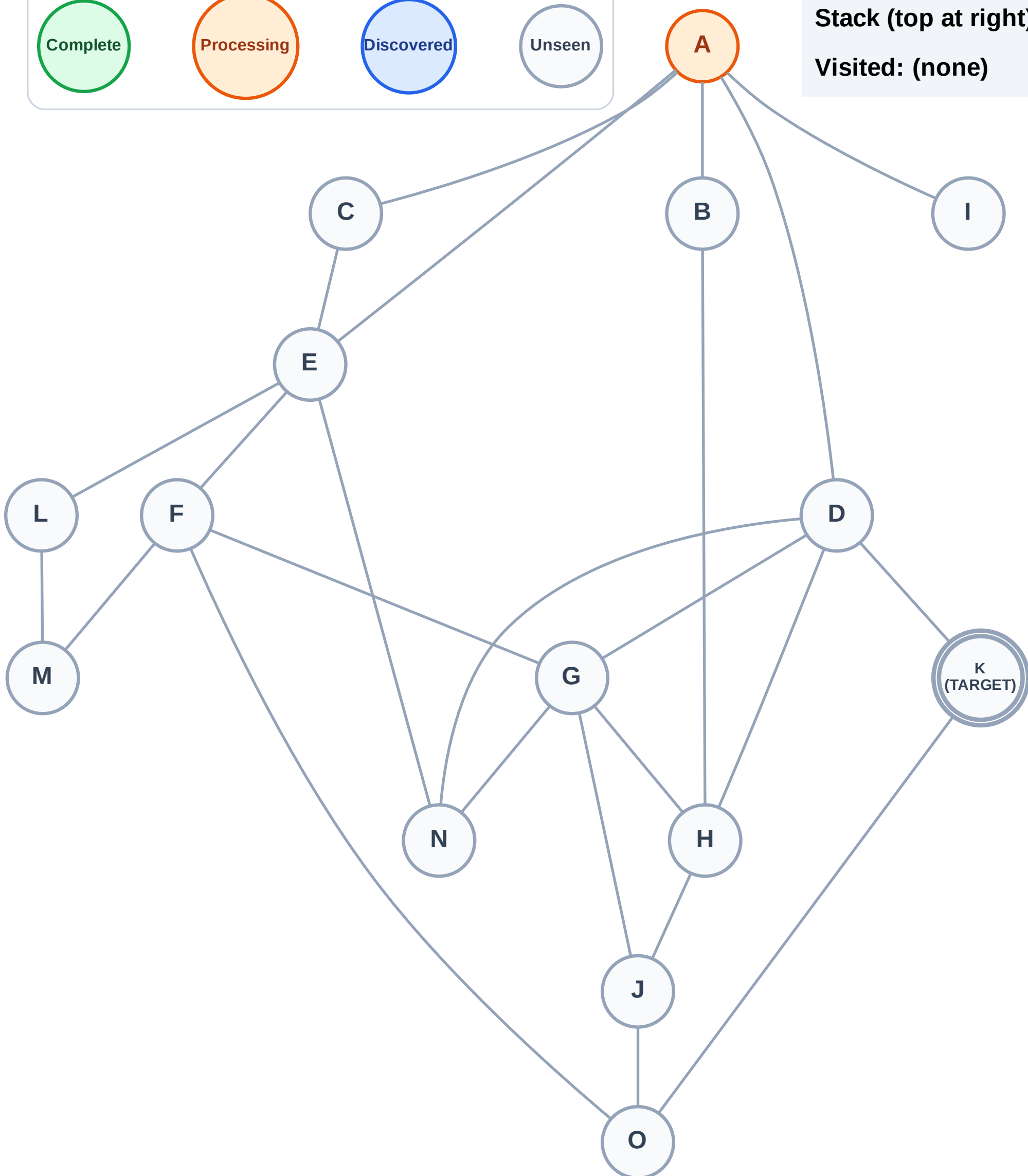
Step 2: Process node A

Legend



Stack (top at right): (empty)

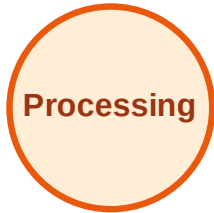
Visited: (none)



DFS Traversal

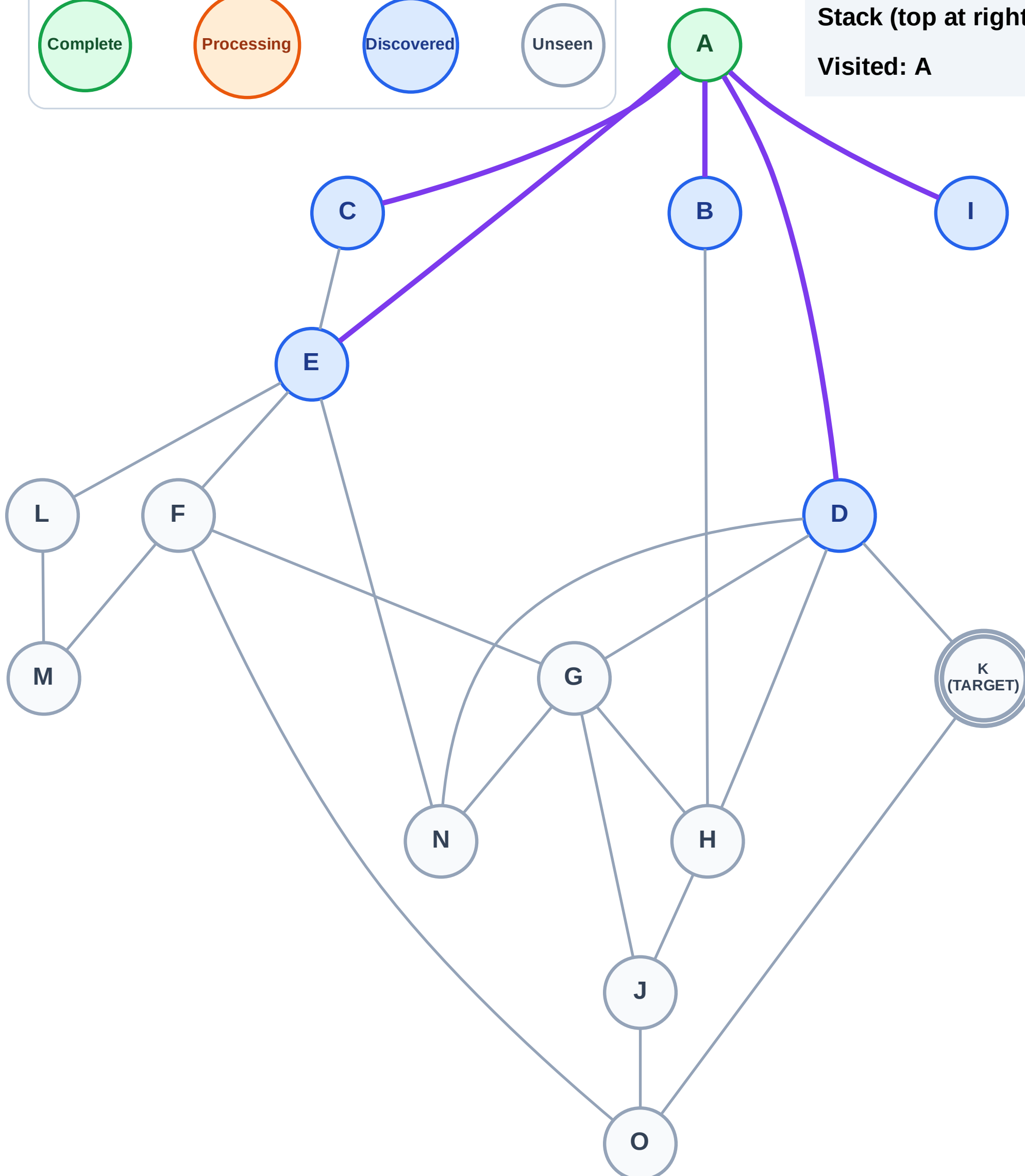
Step 3: Discover I, E, D, C, B from A

Legend



Stack (top at right): I | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

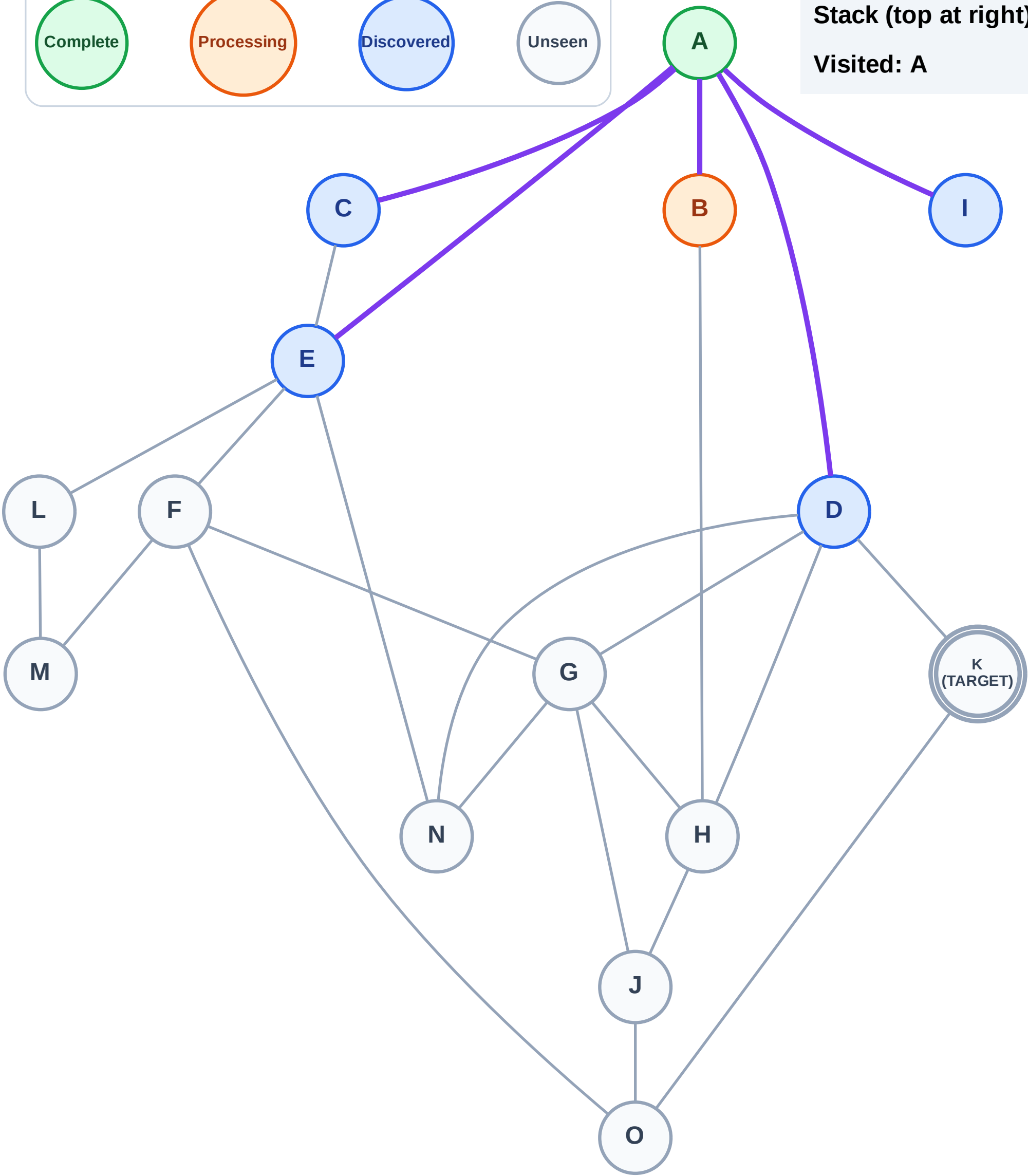
Processing

Discovered

Unseen

Stack (top at right): I | E | D | C

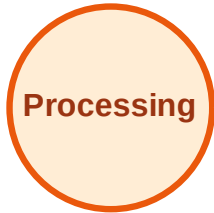
Visited: A



DFS Traversal

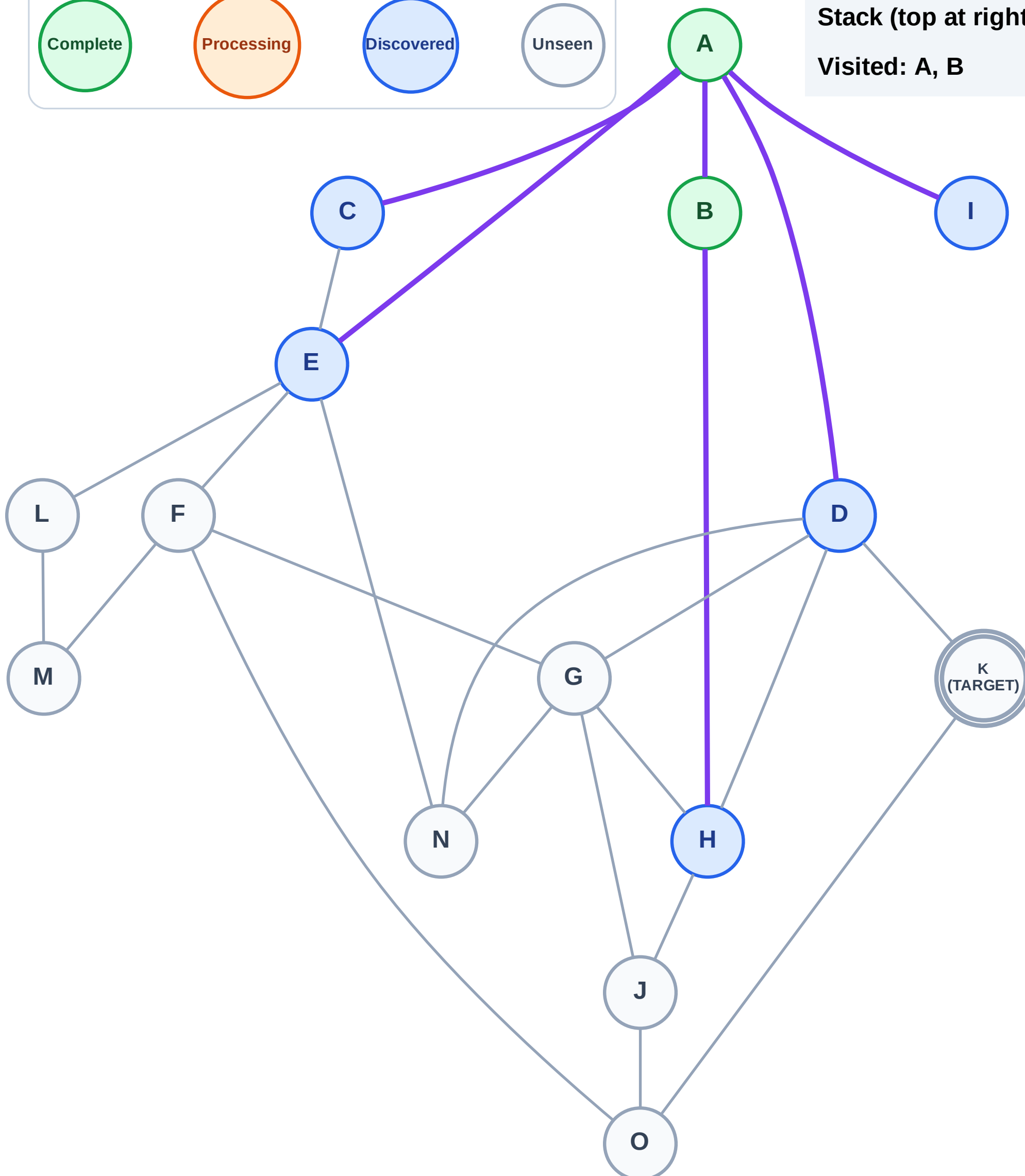
Step 5: Discover H from B

Legend



Stack (top at right): I | E | D | C | H

Visited: A, B



DFS Traversal

Step 6: Process node H

Legend

Complete

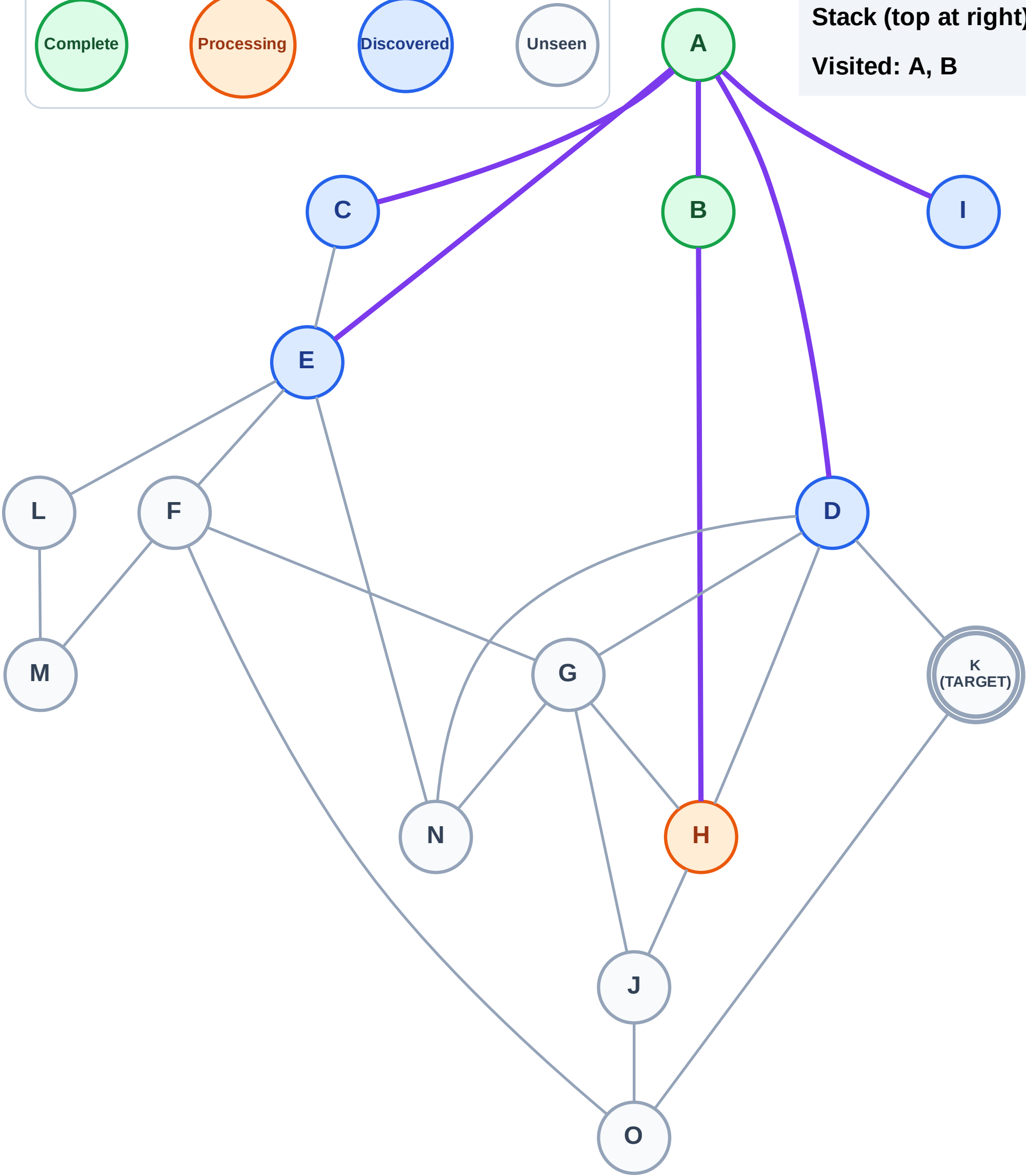
Processing

Discovered

Unseen

Stack (top at right): I | E | D | C

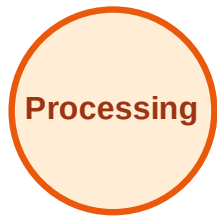
Visited: A, B



DFS Traversal

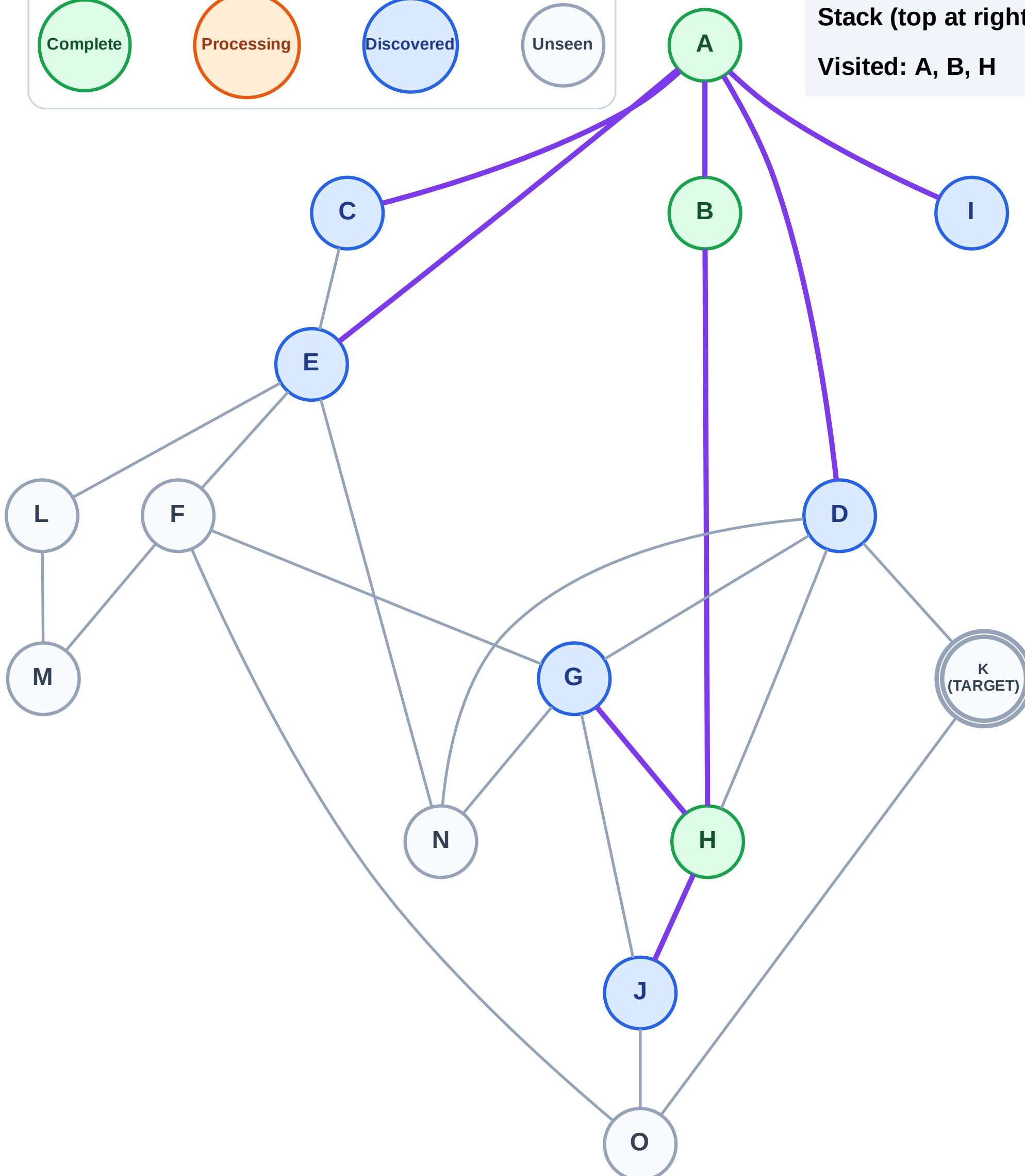
Step 7: Discover J, G from H

Legend



Stack (top at right): I | E | D | C | J | G

Visited: A, B, H



DFS Traversal

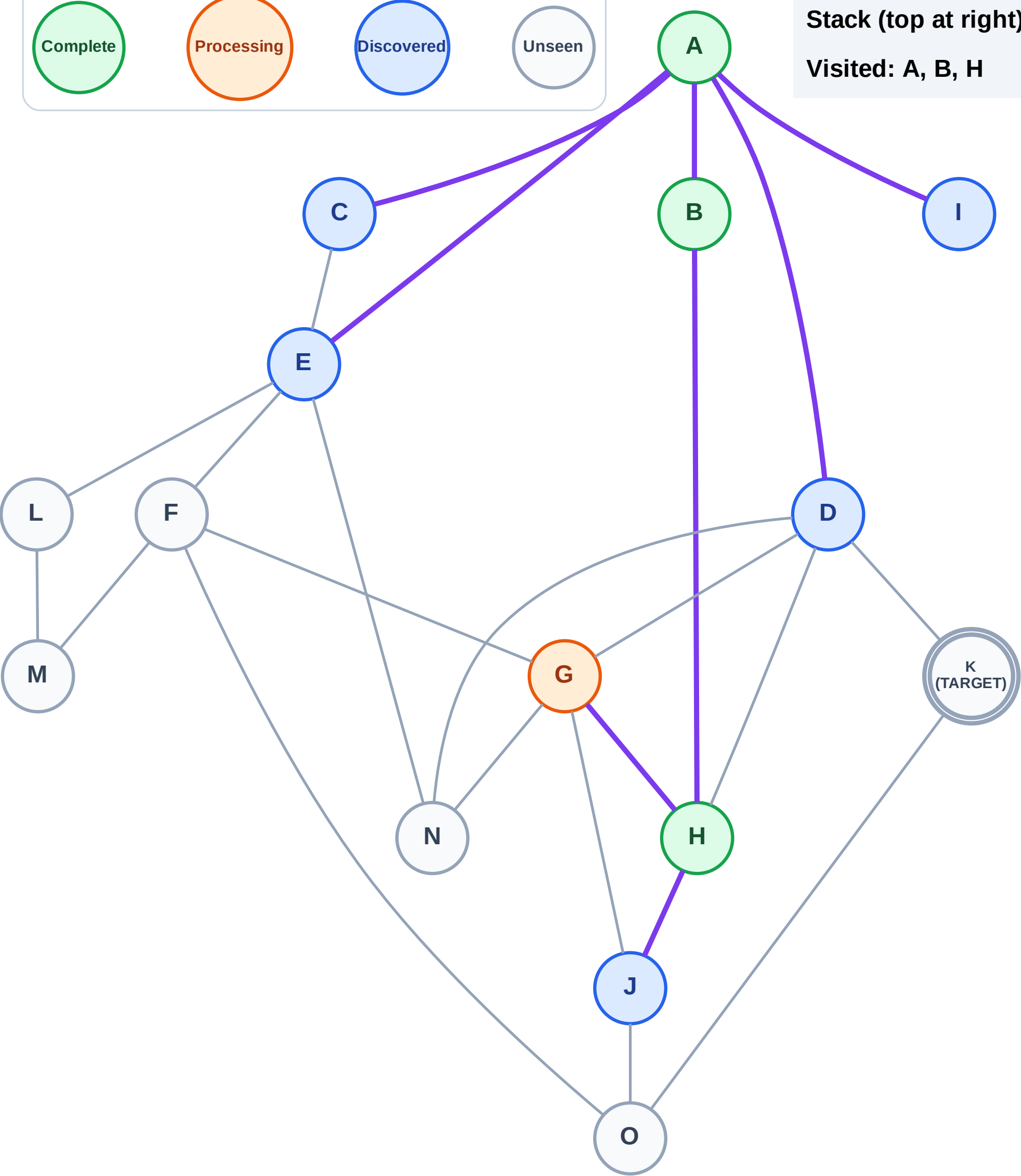
Step 8: Process node G

Legend



Stack (top at right): I | E | D | C | J

Visited: A, B, H



DFS Traversal

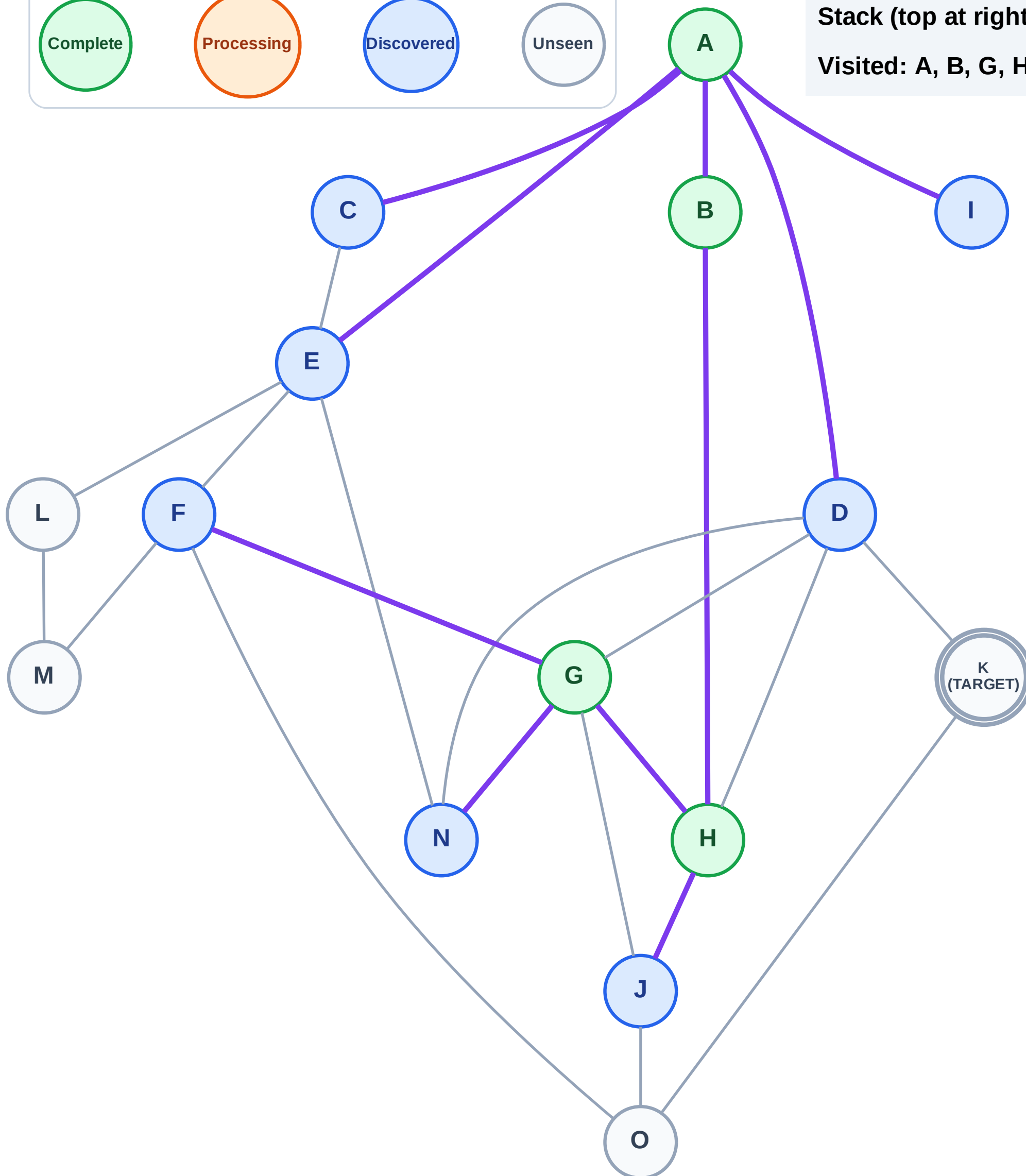
Step 9: Discover N, F from G

Legend



Stack (top at right): I | E | D | C | J | N | F

Visited: A, B, G, H



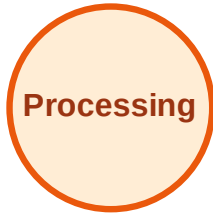
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



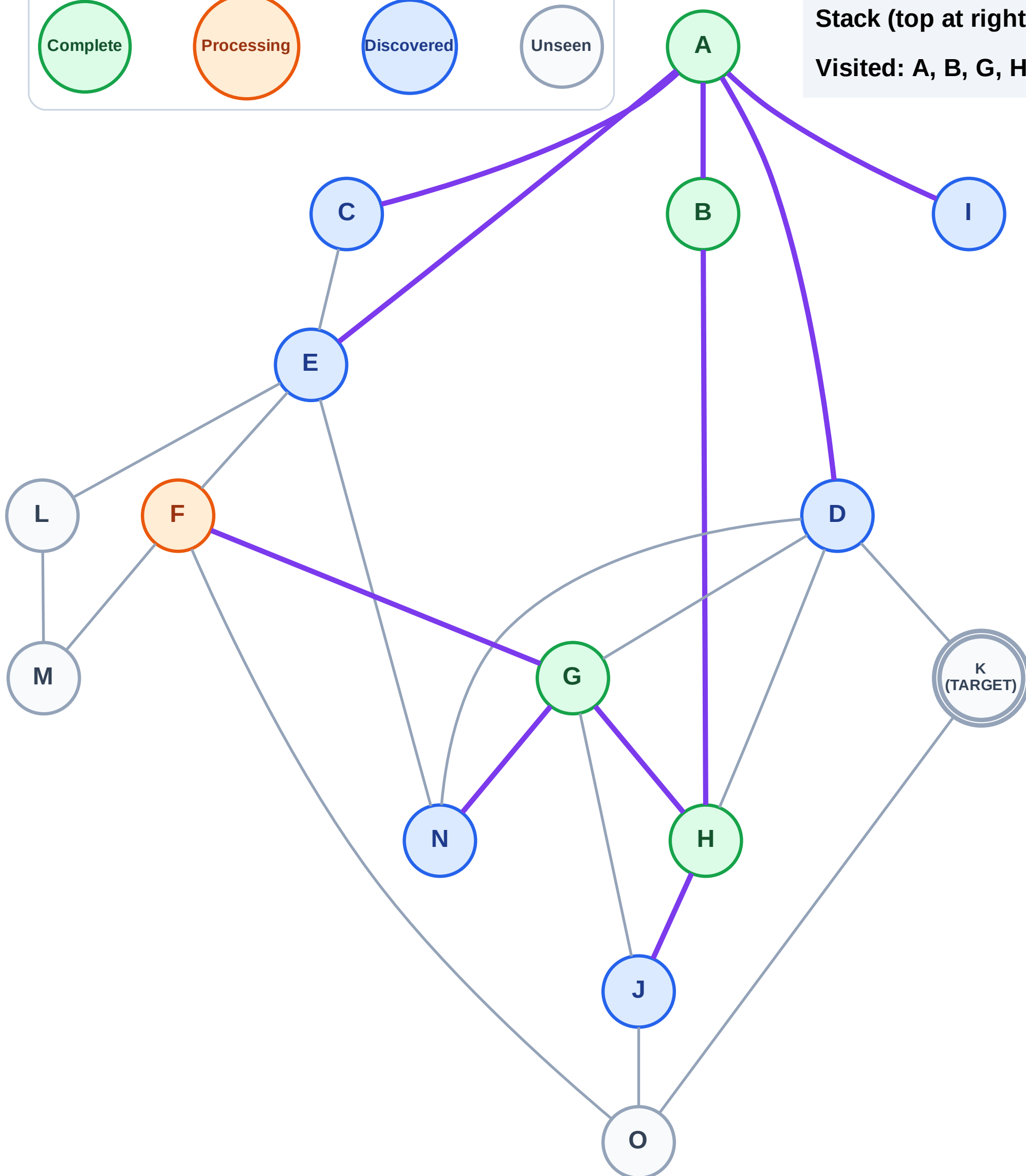
Discovered



Unseen

Stack (top at right): I | E | D | C | J | N

Visited: A, B, G, H



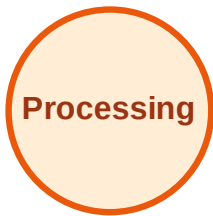
DFS Traversal

Step 11: Discover O, M from F

Legend



Complete



Processing



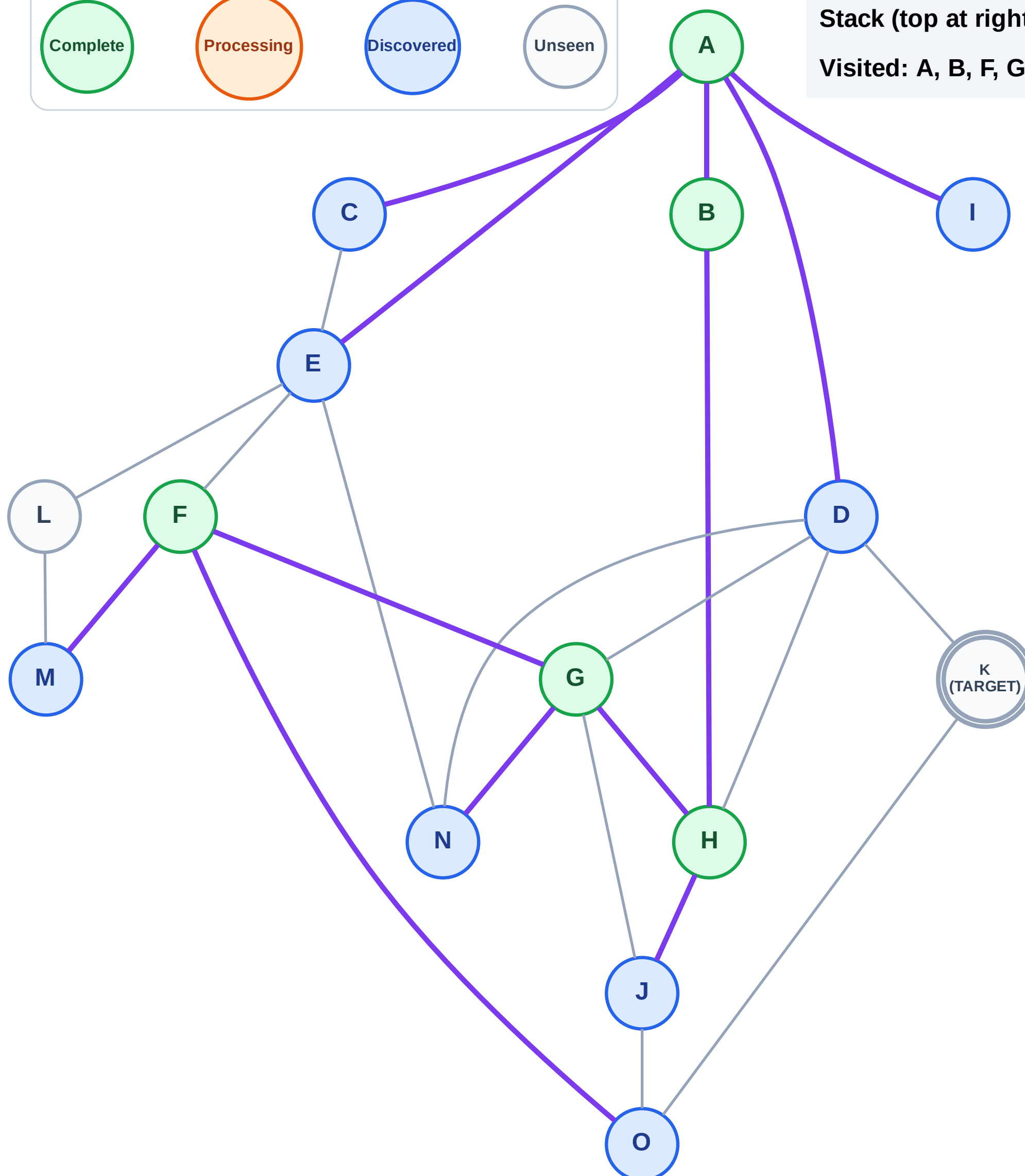
Discovered



Unseen

Stack (top at right): I | E | D | C | J | N | O | M

Visited: A, B, F, G, H



DFS Traversal

Step 12: Process node M

Legend



Complete



Processing



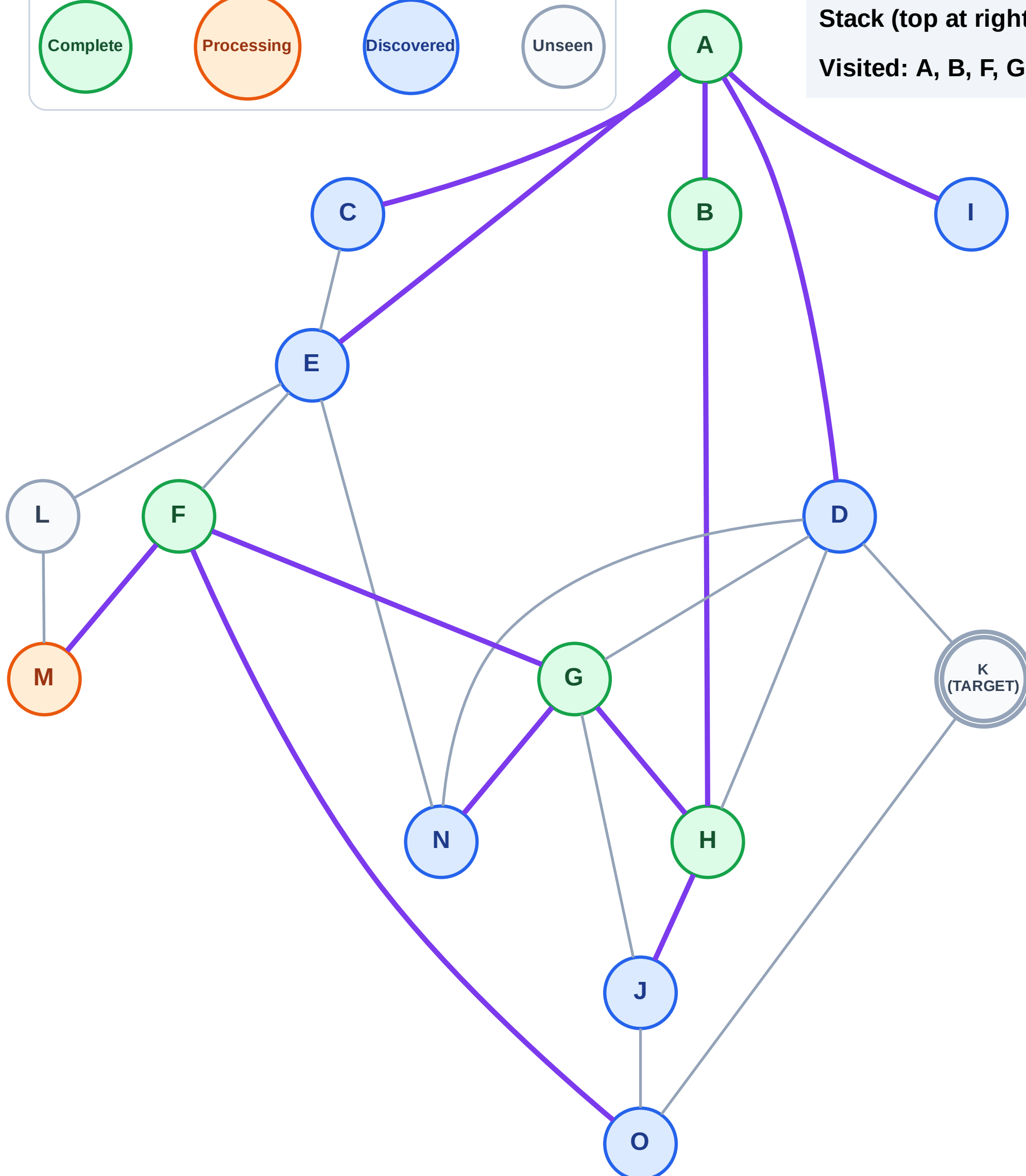
Discovered



Unseen

Stack (top at right): I | E | D | C | J | N | O

Visited: A, B, F, G, H



DFS Traversal

Step 13: Discover L from M

Legend

Complete

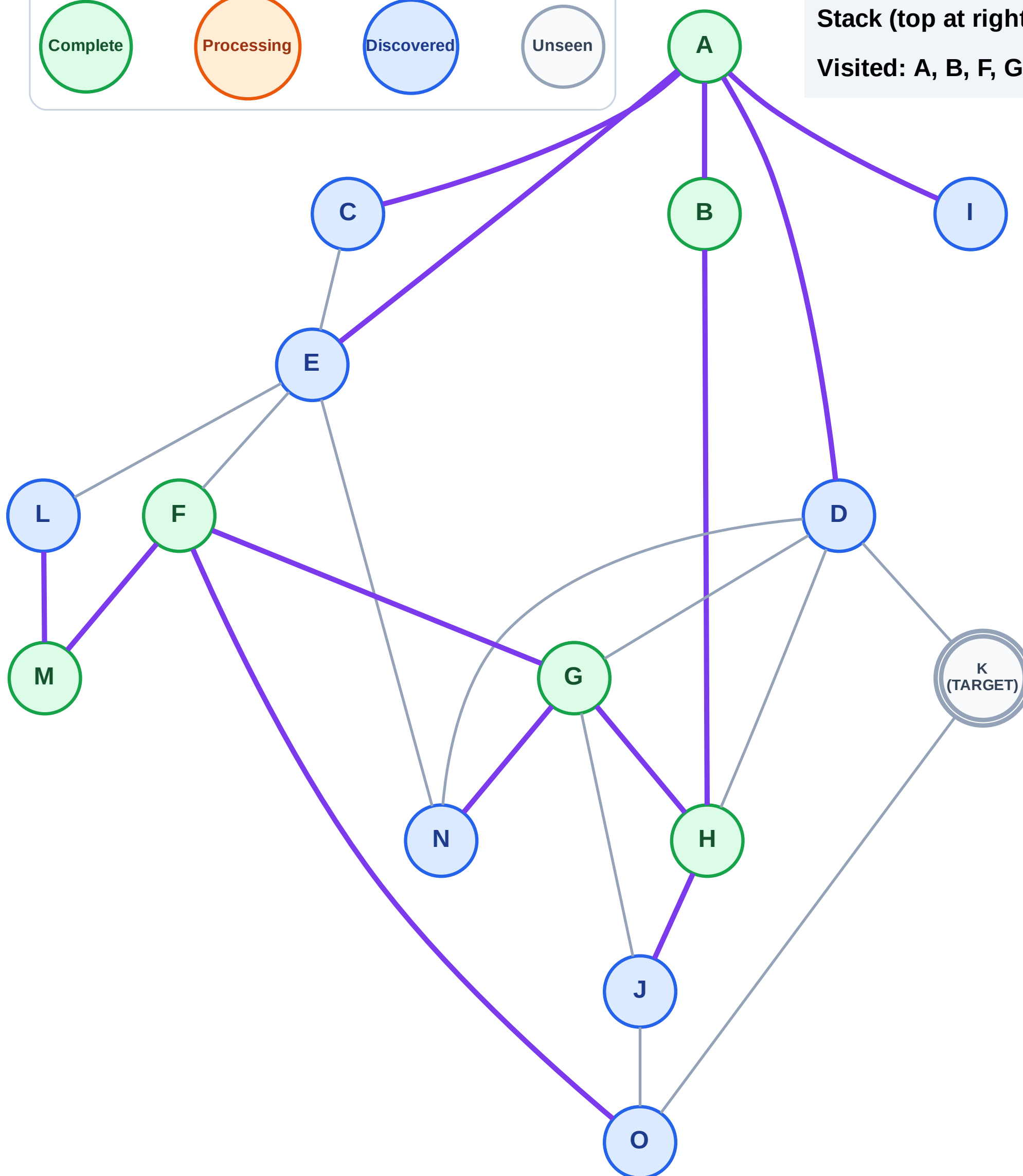
Processing

Discovered

Unseen

Stack (top at right): I | E | D | C | J | N | O | L

Visited: A, B, F, G, H, M



DFS Traversal

Step 14: Process node L

Legend

Complete

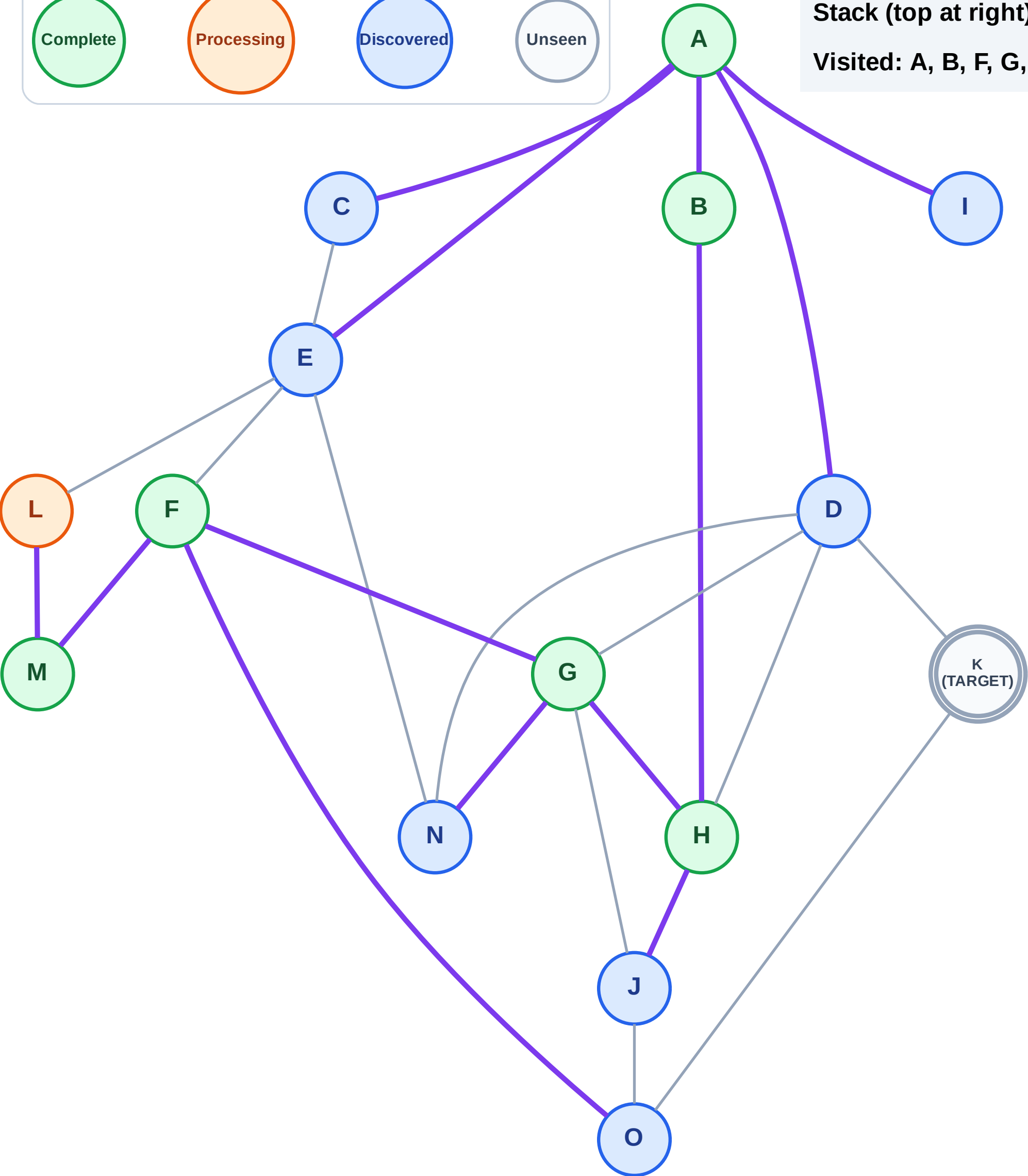
Processing

Discovered

Unseen

Stack (top at right): I | E | D | C | J | N | O

Visited: A, B, F, G, H, M



DFS Traversal

Step 15: No new neighbors from L

Legend

Complete

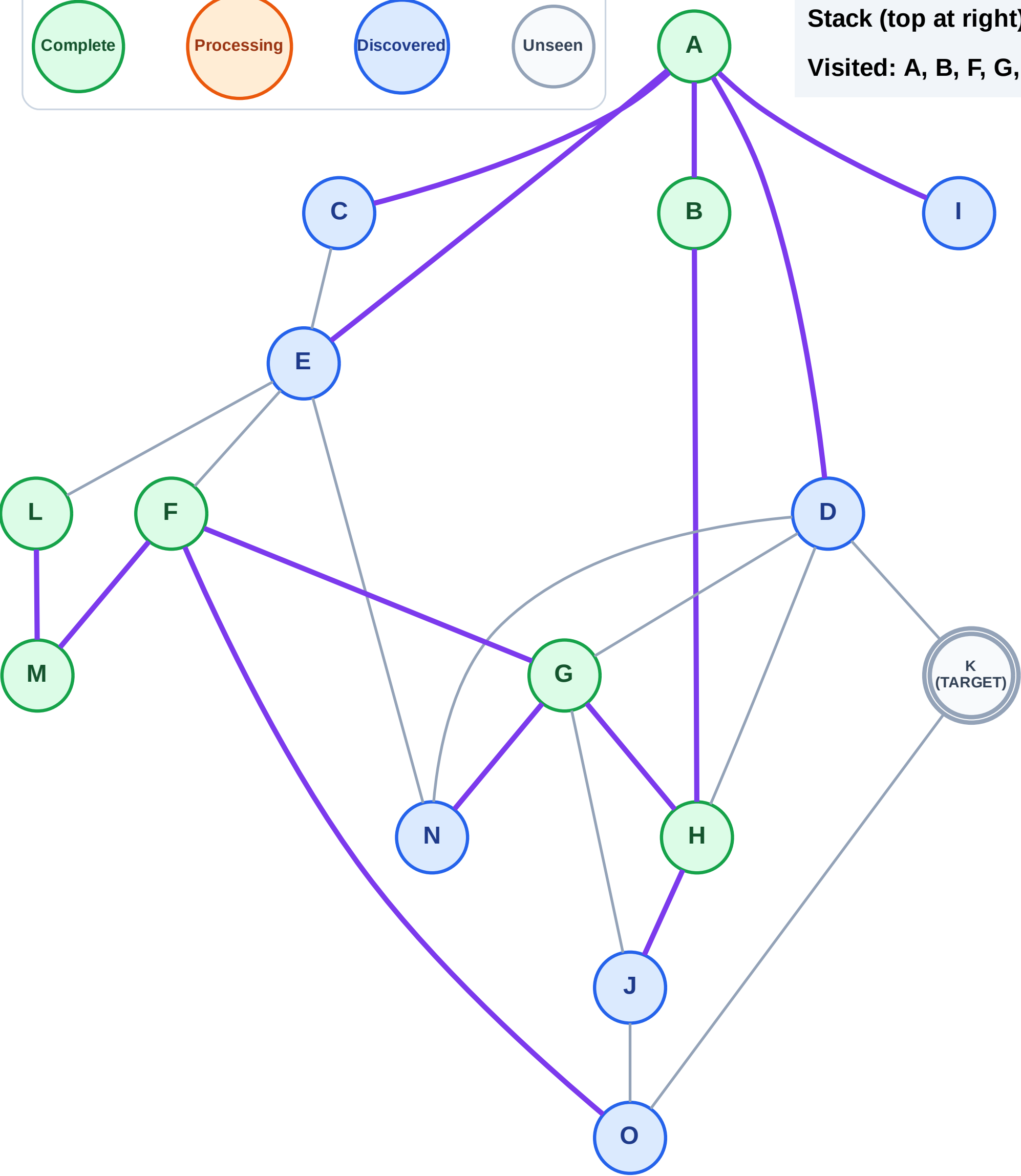
Processing

Discovered

Unseen

Stack (top at right): I | E | D | C | J | N | O

Visited: A, B, F, G, H, L, M



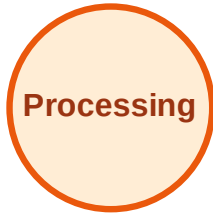
DFS Traversal

Step 16: Process node O

Legend



Complete



Processing



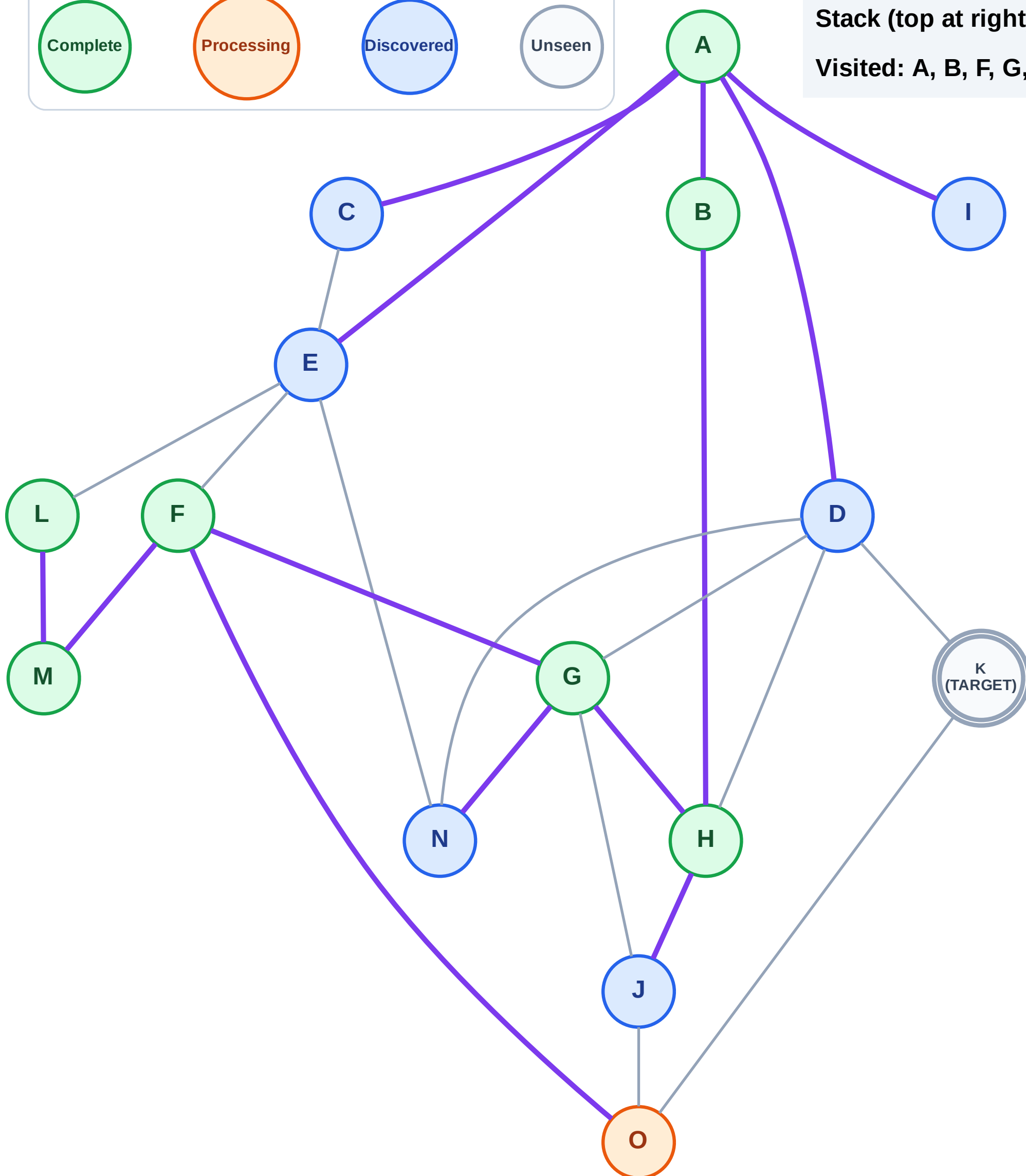
Discovered



Unseen

Stack (top at right): I | E | D | C | J | N

Visited: A, B, F, G, H, L, M



DFS Traversal

Step 17: Discover K from O

Legend

Complete

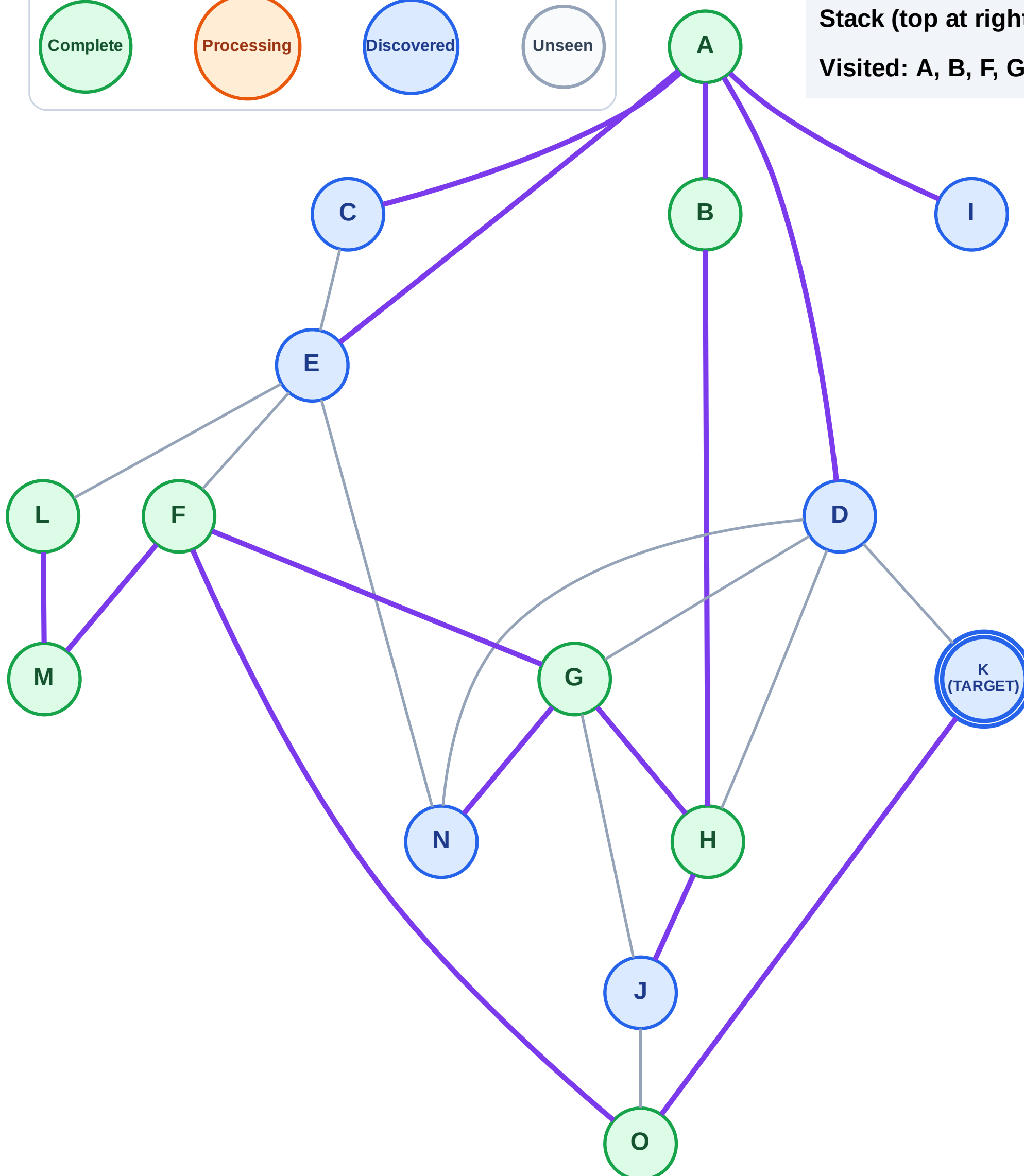
Processing

Discovered

Unseen

Stack (top at right): I | E | D | C | J | N | K

Visited: A, B, F, G, H, L, M, O



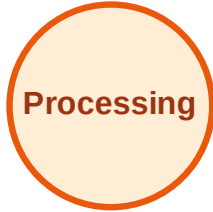
DFS Traversal

Step 18: Process node K

Legend



Complete



Processing



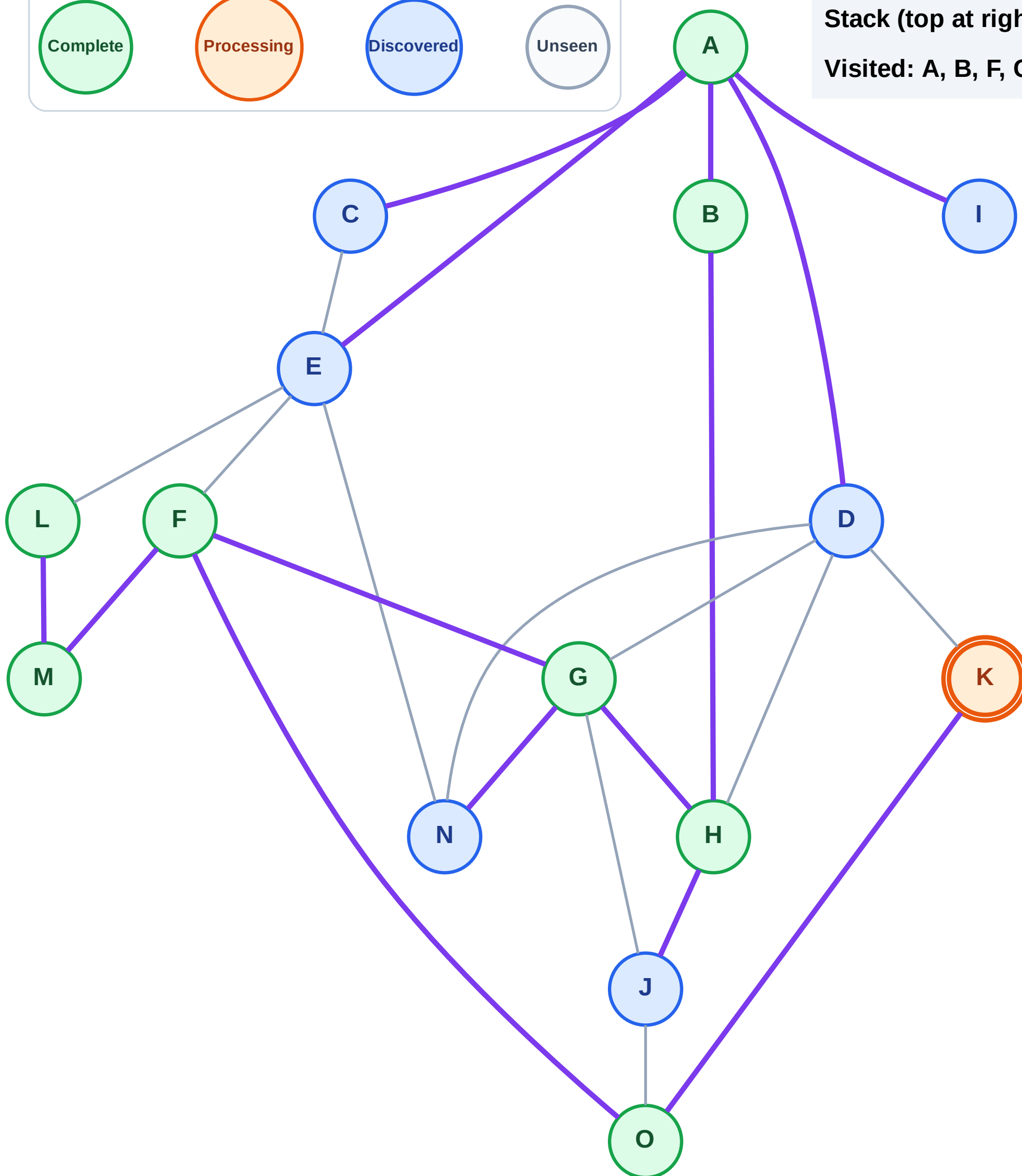
Discovered



Unseen

Stack (top at right): I | E | D | C | J | N

Visited: A, B, F, G, H, L, M, O



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (6 hops): A → B → H → G → F → O → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): I | E | D | C | J | N

Visited: A, B, F, G, H, K, L, M, O

